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TD N° 5: Administration Systèmes Informatiques

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Objectif

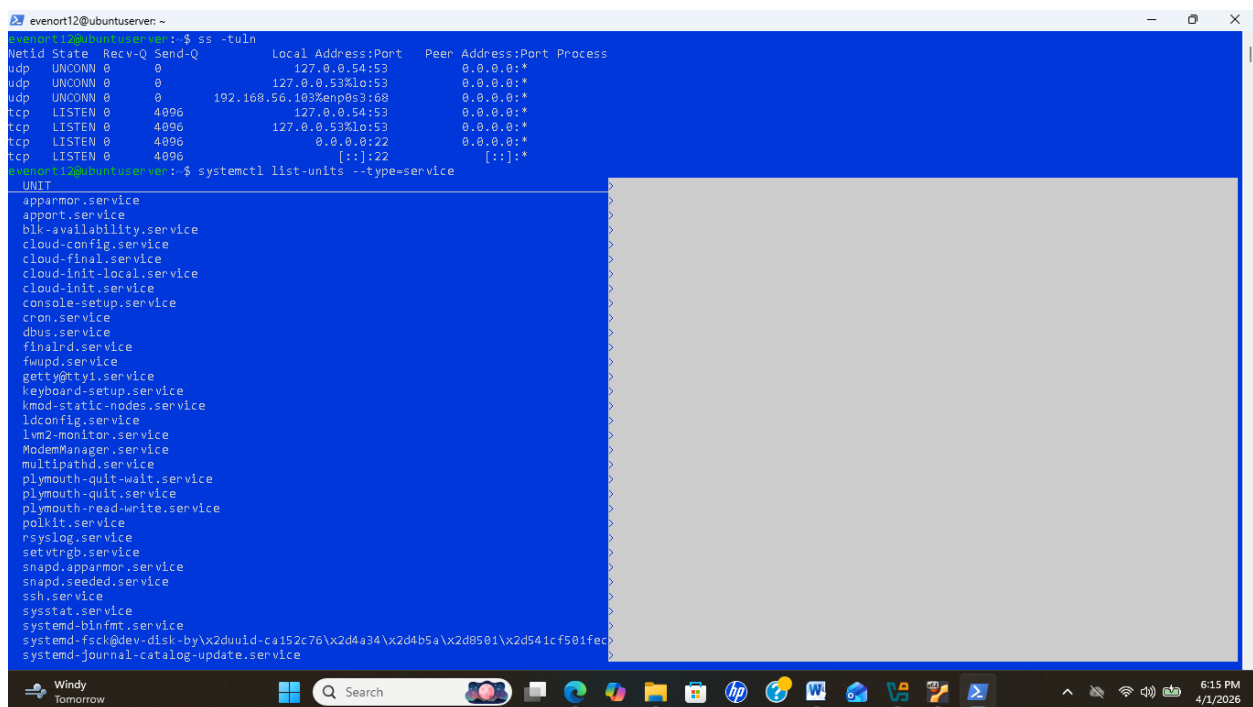
L'objectif de ce travail est d'analyser la sécurité d'un serveur Ubuntu en utilisant l'outil d'audit Lynis, afin d'identifier les vulnérabilités potentielles et d'appliquer des mesures de sécurité pour protéger le serveur contre les attaques. Ce travail permet également de comprendre l'importance des mises à jour, des permissions des fichiers, des systèmes de détection d'intrusion et des audits de sécurité dans un environnement de production.

1. Analyse initiale du serveur

1.1 Vérifier les ports ouverts :

1.2 ss -tuln

1.3 Vérifier les services actifs systemctl list-units --type=service



```
evenort12@ubuntu:~$ ss -tuln
State     Recv-Q    Send-Q     Local Address:Port   Peer Address:Port   Process
udp        UNCONN    0        127.0.0.54:53        0.0.0.0:*            systemd-resolved
udp        UNCONN    0        192.168.56.103:53    0.0.0.0:*            dnsmasq
tcp        LISTEN    0        127.0.0.54:53        0.0.0.0:*            systemd-resolved
tcp        LISTEN    0        127.0.0.53:53        0.0.0.0:*            sshd
tcp        LISTEN    0        0.0.0.0:22           0.0.0.0:*            sshd
tcp        LISTEN    0        :::22               :::*                  sshd

evenort12@ubuntu:~$ systemctl list-units --type=service
UNIT
apparmor.service
apport.service
blk-availability.service
cloud-config.service
cloud-final.service
cloud-init-local.service
cloud-init.service
console-setup.service
cron.service
dbus.service
finalrd.service
fwupd.service
getty@tty1.service
keyboard-setup.service
kmod-static-nodes.service
ldconfig.service
lvm2-monitor.service
ModemManager.service
multipathd.service
plymouth-quit-wait.service
plymouth-quit.service
plymouth-read-write.service
polkit.service
rsyslog.service
setvtrngb.service
snapd.apparmor.service
snapd.seeded.service
ssh.service
sysstat.service
systemd-binfmt.service
systemd-fsck@dev-disk-by\x2duuid-ca152c76\x2d4a34\x2d4b5a\x2d8501\x2d541cf501fec
systemd-journal-catalog-update.service
```

```

evenort12@ubuntu:~$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
apparmor.service                   loaded active exited Load AppArmor profiles
apport.service                      loaded active exited automatic crash report generation
blk-availability.service            loaded active exited Availability of block devices
cloud-config.service               loaded active exited Cloud-init: Config Stage
cloud-final.service                loaded active exited Cloud-init: Final Stage
cloud-init-local.service           loaded active exited Cloud-init: Local Stage (pre-network)
cloud-init.service                 loaded active exited Cloud-init: Network Stage
console-setup.service              loaded active exited Set console font and keymap
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
finalrd.service                    loaded active exited Create final runtime dir for shutdown pivot root
fwupd.service                      loaded active running Firmware update daemon
getty@tty1.service                 loaded active running Getty on tty1
keyboard-setup.service              loaded active exited Set the console keyboard layout
kmod-static-nodes.service           loaded active exited Create list of Static Device Nodes
ldconfig.service                   loaded active exited Rebuild Dynamic Linker Cache
lvm2-monitor.service               loaded active exited Monitoring of LVM2 mirrors, snapshots etc. using dmev
ModemManager.service               loaded active running Modem Manager
multipathd.service                 loaded active running Device-Mapper Multipath Device Controller
plymouth-quit-wait.service          loaded active exited Hold until boot process finishes up
plymouth-quit.service              loaded active exited Terminate Plymouth Boot Screen
plymouth-read-write.service         loaded active exited Tell Plymouth To Write Out Runtime Data
polkit.service                     loaded active running Authorization Manager
rsyslog.service                    loaded active running System Logging Service
setvtrgb.service                   loaded active exited Set console scheme
snapd.apparmor.service              loaded active exited Load AppArmor profiles managed internally by snapd
snapd.seeded.service               loaded active exited Wait until snapd is fully seeded
ssh.service                        loaded active running OpenBSD Secure Shell server
sysstat.service                    loaded active exited Resets System Activity Logs
systemd-binfmt.service              loaded active exited Set Up Additional Binary Formats
systemd-fsck@dev-disk-by\x2duuid-ca152c76\x2d4a34\x2d4b5a\x2d8501\x2d541cf501fecb.service loaded active exited File System Check on /dev/disk/by-uuid/ca152c76-4a34-

```

Figure 1, 2: Les figures montrent la vérification des ports ouverts avec `ss -tln` et l'état des services avec `systemctl list-units --type=service`. Ces commandes permettent de contrôler que les services fonctionnent correctement et que les ports nécessaires sont actifs (voir pages 2 à 3).

1.4 Vérifier les utilisateurs : cat /etc/passwd

```
evenort12@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail list Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
pollinate:x:991:991:User for pollinate:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:146:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
evenort12:x:1000:1000:Marc-Evenort:/home/evenort12:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
evenort12@ubuntu:~$
```

Figure 3: Sur cette image, la commande `cat /etc/passwd` a été utilisée pour vérifier les différents utilisateurs du système Linux.

2. Sécurisation de SSH

2.1 Éditer : `sudo nano /etc/ssh/sshd_config`

Appliquer les modifications suivantes :

`PermitRootLogin no`

`PasswordAuthentication yes`

`Port 2222`

`MaxAuthTries 3`

`AllowUsers admin evenort12`

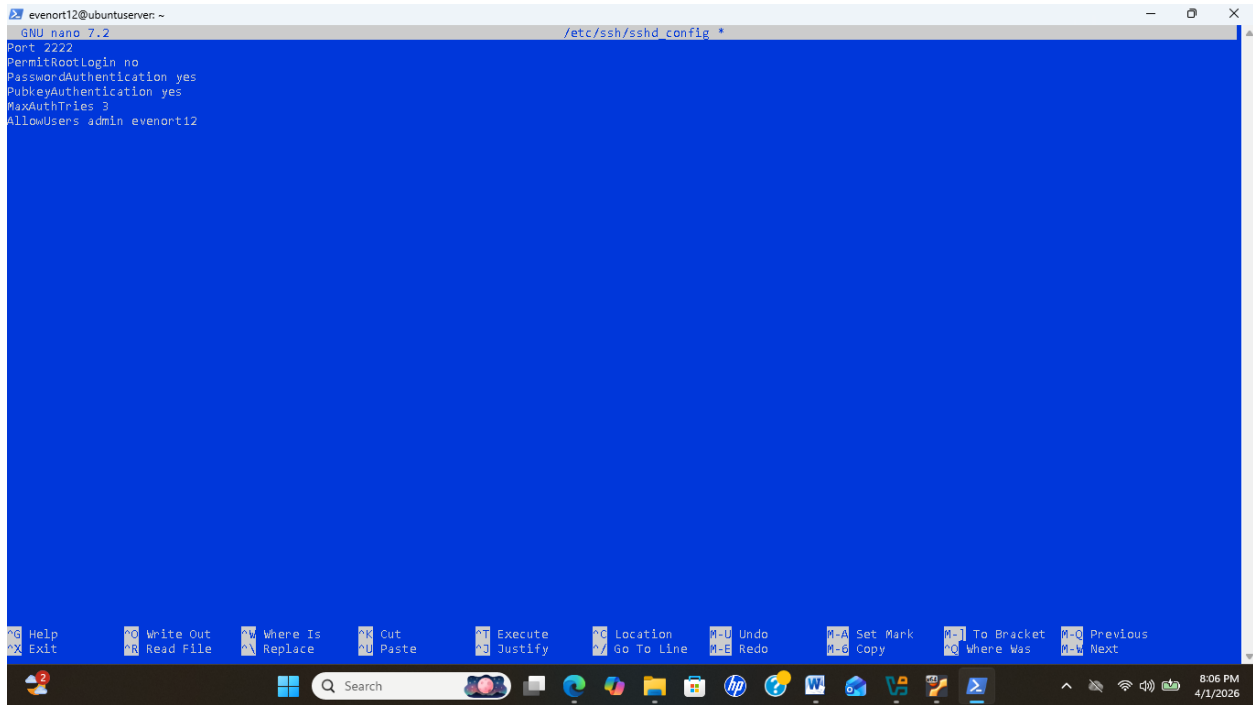


Figure 4: Cette image montre les différentes permissions accordées afin de sécuriser le port SSH.

2.2 Redémarrer le service :

```
sudo systemctl restart ssh
```

2.3 Tester la connexion :

```
ssh -p 2222 evenort@192.168.56.103
```

3. Configuration du Firewall

3.1 Installer : `sudo apt install ufw -y`

3.2 Configurer lasudosu politique par défaut :

```
sudo ufw default deny incoming  
sudo ufw default allow outgoing
```

3.3 Autoriser :

```
Sudo ufw allow 2222/tcp  
Sudo ufw allow 80/tcp  
Sudo ufw allow 443tcp
```

3.4 Activer le firewall :

sudo ufw enable
3.5 Vérifier :
sudo ufw status verbose

```
Ubuntu_Evenort5 [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

evenort12@ubuntu:~$ sudo apt install ufw -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
ufw est déjà la version la plus récente (0.36.2-6).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 167 non mis à jour.
evenort12@ubuntu:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
evenort12@ubuntu:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
evenort12@ubuntu:~$ sudo ufw status
Status: active

To                Action      From
-----
2222/tcp           ALLOW      Anywhere
2222/tcp (v6)      ALLOW      Anywhere (v6)

evenort12@ubuntu:~$ sudo ufw allow 2222/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
evenort12@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
evenort12@ubuntu:~$ sudo ufw allow 443/tcp
Rule added
Rule added (v6)
evenort12@ubuntu:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
evenort12@ubuntu:~$ sudo ufw status
Status: active

To                Action      From
-----
2222/tcp           ALLOW      Anywhere
80/tcp            ALLOW      Anywhere
443/tcp           ALLOW      Anywhere
```

```
Ubuntu_Evenort5 [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

evenort12@ubuntu:~$ sudo ufw allow 2222/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
evenort12@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
evenort12@ubuntu:~$ sudo ufw allow 443/tcp
Rule added
Rule added (v6)
evenort12@ubuntu:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
evenort12@ubuntu:~$ sudo ufw status
Status: active

To                Action      From
-----
2222/tcp           ALLOW      Anywhere
80/tcp            ALLOW      Anywhere
443/tcp           ALLOW      Anywhere
2222/tcp (v6)     ALLOW      Anywhere (v6)
80/tcp (v6)       ALLOW      Anywhere (v6)
443/tcp (v6)      ALLOW      Anywhere (v6)

evenort12@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                Action      From
-----
2222/tcp           ALLOW IN    Anywhere
80/tcp            ALLOW IN    Anywhere
443/tcp           ALLOW IN    Anywhere
2222/tcp (v6)     ALLOW IN    Anywhere (v6)
80/tcp (v6)       ALLOW IN    Anywhere (v6)
443/tcp (v6)      ALLOW IN    Anywhere (v6)

evenort12@ubuntu:~$
```

Figure 5, 6: Ces images illustrent l'installation du pare-feu, l'autorisation des ports et l'activation du pare-feu afin de vérifier le fonctionnement du service en mode verbose sur le serveur (voir pages 5 à 6).

4. Protection contre brute force

4.1 Installer : `sudo apt install fail2ban -y`

4.2 Créer la configuration : `sudo nano /etc/fail2ban/jail.local`

Ajout :

[sshd]

enabled = true

port = 2222

maxretry = 3

bantime = 600

4.3 Redémarrer : `sudo systemctl restart fail2ban`

4.4 Vérifier : `sudo fail2ban-client status`

5 Analyse des logs

5.1 Afficher les logs SSH : `cat /var/log/auth.log`

5.2 Filtrer les erreurs : `grep "Failed password" /var/log/auth.log`

5.3 Afficher les logs Fail2Ban : `cat /var/log/fail2ban.log`

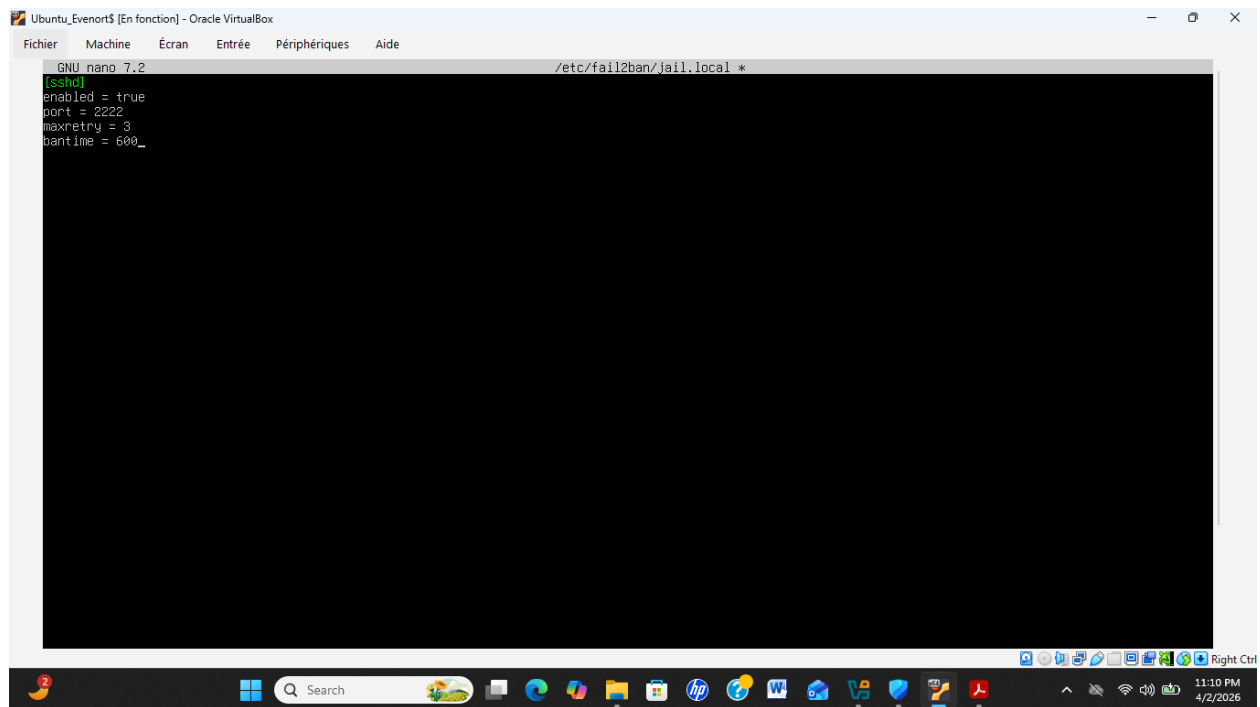


Figure 7: Après avoir installé Fail2ban, j'ai configuré le port SSH afin de protéger le serveur contre les attaques par force brute.


```
evenort12@ubuntu:~$ sudo fail2ban-client status
[sudo] password for evenort12:
Status
|- Number of fail: 1
|- Fail list: sshd
evenort12@ubuntu:~$ cat /var/log/auth.log
2026-04-01T20:27:10.443456+00:00 ubuntu: useradd[862]: new group: name=evenort12, GID=1000
2026-04-01T20:27:10.443468+00:00 ubuntu: useradd[862]: new user: name=evenort12, UID=1000, GID=1000, home=/home/evenort12, shell=/bin/bash, from=none
2026-04-01T20:27:10.443473+00:00 ubuntu: useradd[862]: add 'evenort12' to group 'adm'
2026-04-01T20:27:10.443478+00:00 ubuntu: useradd[862]: add 'evenort12' to group 'cdrom'
2026-04-01T20:27:10.443480+00:00 ubuntu: useradd[862]: add 'evenort12' to group 'sudo'
2026-04-01T20:27:10.443484+00:00 ubuntu: useradd[862]: add 'evenort12' to group 'dip'
2026-04-01T20:27:10.443487+00:00 ubuntu: useradd[862]: add 'evenort12' to group 'plugdev'
2026-04-01T20:27:10.443489+00:00 ubuntu: useradd[862]: add 'evenort12' to group 'lxd'
2026-04-01T20:27:10.443492+00:00 ubuntu: useradd[862]: add 'evenort12' to shadow group 'adm'
2026-04-01T20:27:10.443503+00:00 ubuntu: useradd[862]: add 'evenort12' to shadow group 'cdrom'
2026-04-01T20:27:10.443507+00:00 ubuntu: useradd[862]: add 'evenort12' to shadow group 'sudo'
2026-04-01T20:27:10.443510+00:00 ubuntu: useradd[862]: add 'evenort12' to shadow group 'dip'
2026-04-01T20:27:10.443513+00:00 ubuntu: useradd[862]: add 'evenort12' to shadow group 'plugdev'
2026-04-01T20:27:10.443514+00:00 ubuntu: useradd[862]: add 'evenort12' to shadow group 'lxd'
2026-04-01T20:27:10.444141+00:00 ubuntu: passwd[870]: password for 'evenort12' changed by 'root'
2026-04-01T20:27:10.444145+00:00 ubuntu: systemd-logind[001]: New seat seat0.
2026-04-01T20:27:10.444148+00:00 ubuntu: systemd-logind[001]: Watching system buttons on /dev/input/event0 (Power Button)
2026-04-01T20:27:10.444151+00:00 ubuntu: systemd-logind[001]: Watching system buttons on /dev/input/event1 (Sleep Button)
2026-04-01T20:27:10.298208+00:00 ubuntu: polkitd[890]: Loading rules from directory /etc/polkit-1/rules.d
2026-04-01T20:27:10.302288+00:00 ubuntu: polkitd[890]: Loading rules from directory /usr/share/polkit-1/rules.d
2026-04-01T20:27:10.386014+00:00 ubuntu: polkitd[890]: Finished loading, compiling and executing 6 rules
2026-04-01T20:27:10.410476+00:00 ubuntu: polkitd[890]: Acquired the name org.freedesktop.PolicyKit1 on the system bus
2026-04-01T20:28:08.656609+00:00 ubuntu: login[988]: PAM unable to dlopen(pam_lastlog.so): /usr/lib/security/pam_lastlog.so: cannot open shared object file: No such file or directory
2026-04-01T20:28:08.656821+00:00 ubuntu: login[988]: PAM adding faulty module: pam_lastlog.so
2026-04-01T20:28:16.783958+00:00 ubuntu: login[988]: pam_unix(login:session): session opened for user evenort12(uid=1000) by evenort12(uid=0)
2026-04-01T20:28:16.808473+00:00 ubuntu: systemd-logind[001]: New session 1 of user evenort12.
2026-04-01T20:28:16.926371+00:00 ubuntu: (systemd): pam_unix(systemd-user:session): session opened for user evenort12(uid=1000) by evenort12(uid=0)
2026-04-01T20:31:34.950109+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/apt update
2026-04-01T20:31:34.962871+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T20:31:44.510600+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T20:32:00.349284+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/apt update
2026-04-01T20:32:00.377216+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T20:35:01.299669+00:00 ubuntu: CRON[1485]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T20:35:01.311194+00:00 ubuntu: CRON[1485]: pam_unix(cron:session): session closed for user root
2026-04-01T20:39:22.878508+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
```

```
evenort12@ubuntu:~$ sudo fail2ban-client status
[sudo] password for evenort12:
Status
|- Number of fail: 1
|- Fail list: sshd
evenort12@ubuntu:~$ cat /var/log/auth.log
2026-04-01T20:39:22.878508+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T20:40:30.333937+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/apt upgrade -y
2026-04-01T20:40:30.335854+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T20:45:01.971011+00:00 ubuntu: CRON[1599]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T20:45:01.982737+00:00 ubuntu: CRON[1599]: pam_unix(cron:session): session closed for user root
2026-04-01T20:55:01.060826+00:00 ubuntu: CRON[1609]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T20:55:01.099935+00:00 ubuntu: CRON[1609]: pam_unix(cron:session): session closed for user root
2026-04-01T21:05:01.221326+00:00 ubuntu: CRON[1620]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T21:05:01.237079+00:00 ubuntu: CRON[1620]: pam_unix(cron:session): session closed for user root
2026-04-01T21:15:57.343709+00:00 ubuntu: CRON[1657]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T21:15:57.360964+00:00 ubuntu: CRON[1657]: pam_unix(cron:session): session closed for user root
2026-04-01T21:17:01.967761+00:00 ubuntu: CRON[1692]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T21:17:01.996061+00:00 ubuntu: CRON[1692]: pam_unix(cron:session): session closed for user root
2026-04-01T21:25:01.370364+00:00 ubuntu: CRON[1701]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T21:25:01.394873+00:00 ubuntu: CRON[1701]: pam_unix(cron:session): session closed for user root
2026-04-01T21:35:01.891794+00:00 ubuntu: CRON[1712]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T21:35:01.911431+00:00 ubuntu: CRON[1712]: pam_unix(cron:session): session closed for user root
2026-04-01T21:37:05.287760+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T21:37:32.846117+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/apt update
2026-04-01T21:37:32.847120+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T21:40:37.808524+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T21:41:07.917251+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/apt install openssh-server -y
2026-04-01T21:41:07.918496+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T21:41:40.659511+00:00 ubuntu: useradd[2291]: new user: name=sshd, UID=100, GID=65534, home=/run/sshd, shell=/usr/sbin/nologin, from=none
2026-04-01T21:41:40.982650+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T21:42:12.892960+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/systemctl status ssh
2026-04-01T21:42:12.921921+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T21:42:12.947680+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T21:42:28.048154+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/systemctl start ssh
2026-04-01T21:42:28.049511+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T21:42:28.126083+00:00 ubuntu: sshd[2486]: Server listening on 0.0.0.0 port 22.
2026-04-01T21:42:28.127055+00:00 ubuntu: sshd[2486]: Server listening on : port 22.
2026-04-01T21:42:28.131466+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T21:42:51.024904+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/systemctl enable ssh
2026-04-01T21:42:51.026758+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T21:42:52.207795+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T21:42:59.238169+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/systemctl status ssh
2026-04-01T21:42:59.239175+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T21:42:59.306960+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T21:45:02.046870+00:00 ubuntu: CRON[2679]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T21:45:02.806951+00:00 ubuntu: CRON[2679]: pam_unix(cron:session): session closed for user root
2026-04-01T21:46:50.656393+00:00 ubuntu: sudo: evenort12 : TTY=ttty1 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/sbin/ufw allow 22
2026-04-01T21:46:50.656997+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
```

```
evenort12@ubuntu:~$ cat /var/log/auth.log
2026-04-01T12:14:00.293048+00:00 ubuntu: sshd[2724]: pam_unix(sshd:session): session opened for user evenort12(uid=1000) by evenort12(uid=0)
2026-04-01T12:14:00.308138+00:00 ubuntu: systemd-logind[901]: New session 12 of user evenort12.
2026-04-01T12:15:07.797822+00:00 ubuntu: CRON[2844]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T12:15:07.827626+00:00 ubuntu: CRON[2844]: pam_unix(cron:session): session closed for user root
2026-04-01T12:15:55.218961+00:00 ubuntu: sshd[2981]: Connection closed by authenticating user evenort12 192.168.56.103 port 58900 [preauth]
2026-04-01T12:15:55.249984+00:00 ubuntu: sshd[2988]: Connection closed by authenticating user evenort12 192.168.56.103 port 40914 [preauth]
2026-04-01T12:15:57.03.974088+00:00 ubuntu: sshd[2920]: Accepted password for evenort12 from 192.168.56.103 port 40926 ssh2
2026-04-01T12:15:57.03.978420+00:00 ubuntu: sshd[2920]: pam_unix(sshd:session): session opened for user evenort12(uid=1000) by evenort12(uid=0)
2026-04-01T12:15:57.03.980571+00:00 ubuntu: systemd-logind[901]: New session 14 of user evenort12.
2026-04-01T12:15:57.04.690306+00:00 ubuntu: sshd[2977]: Received disconnect from 192.168.56.103 port 40926:11: disconnected by user
2026-04-01T12:15:57.04.728127+00:00 ubuntu: sshd[2977]: Disconnected from user evenort12 192.168.56.103 port 40926
2026-04-01T12:15:57.04.721077+00:00 ubuntu: sshd[2920]: pam_unix(sshd:session): session closed for user evenort12
2026-04-01T12:15:57.04.721439+00:00 ubuntu: systemd-logind[901]: Session 14 logged out. Waiting for processes to exit.
2026-04-01T12:15:57.04.721868+00:00 ubuntu: systemd-logind[901]: Removed session 14.
2026-04-01T12:15:57.04.847918+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/cat /home/evenort12/.ssh/id_rsa.pub
2026-04-01T12:15:57.04.852552+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T12:15:57.04.860828+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T12:15:57.04.869664+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/cat /home/evenort12/.ssh/id_rsa.pub
2026-04-01T12:15:57.04.879817+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T12:15:57.04.889881+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T12:15:57.04.899891+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/chmod 700 /home/evenort12/.ssh
2026-04-01T12:15:57.04.909891+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T12:15:57.04.919891+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T12:15:57.04.929891+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/chmod 600 /home/evenort12/.ssh/authorized
2026-04-01T12:15:57.04.939891+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T12:15:57.04.949891+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T12:15:57.04.959891+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/systemctl restart ssh
2026-04-01T12:15:57.04.969891+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-01T12:15:57.04.979891+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T12:15:57.04.989891+00:00 ubuntu: sshd[2722]: Received signal 15; terminating.
2026-04-01T12:15:57.04.999891+00:00 ubuntu: sshd[3025]: Server listening on 0.0.0.0 port 22.
2026-04-01T12:15:57.05.009891+00:00 ubuntu: sshd[3025]: Server listening on :: port 22.
2026-04-01T12:15:57.05.019891+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root
2026-04-01T12:15:57.05.029891+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-01T12:15:57.05.039891+00:00 ubuntu: CRON[3027]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-01T12:15:57.05.049891+00:00 ubuntu: CRON[3027]: pam_unix(cron:session): session closed for user root
2026-04-01T12:15:57.05.059891+00:00 ubuntu: sshd[3033]: Accepted publickey for evenort12 from 192.168.56.103 port 57018 ssh2: ED25519 SHA256:3y4j1qk/80sz5ihcUSGLxhizv
2026-04-01T12:15:57.05.069891+00:00 ubuntu: sshd[3033]: pam_unix(sshd:session): session opened for user evenort12(uid=1000) by evenort12(uid=0)
2026-04-01T12:15:57.05.079891+00:00 ubuntu: systemd-logind[901]: New session 16 of user evenort12.
2026-04-01T12:15:57.05.089891+00:00 ubuntu: sshd[3101]: Accepted publickey for evenort12 from 192.168.56.103 port 52436 ssh2: ED25519 SHA256:3y4j1qk/80sz5ihcUSGLxhizv
2026-04-01T12:15:57.05.099891+00:00 ubuntu: sshd[3101]: pam_unix(sshd:session): session opened for user evenort12(uid=1000) by evenort12(uid=0)
2026-04-01T12:15:57.05.109891+00:00 ubuntu: systemd-logind[901]: New session 17 of user evenort12.
```

```
evenort12@ubuntu:~$ cat /var/log/auth.log
2026-04-03T02:15:31.750935+00:00 ubuntu: sshd[917]: Server listening on 0.0.0.0 port 2222.
2026-04-03T02:15:31.752348+00:00 ubuntu: sshd[917]: Server listening on :: port 2222.
2026-04-03T02:15:31.752348+00:00 ubuntu: login[924]: PAM unable to dlopen(pam_lastlog.so): /usr/lib/security/pam_lastlog.so: cannot open shared object file: No such
file or directory
2026-04-03T02:15:31.752348+00:00 ubuntu: login[924]: PAM adding faulty module: pam_lastlog.so
2026-04-03T02:15:31.752348+00:00 ubuntu: login[924]: pam_unix(login:session): session opened for user evenort12(uid=1000) by evenort12(uid=0)
2026-04-03T02:15:31.752348+00:00 ubuntu: systemd-logind[778]: New session 1 of user evenort12.
2026-04-03T02:15:31.752348+00:00 ubuntu: (systemd): pam_unix(systemd-user:session): session opened for user evenort12(uid=1000) by evenort12(uid=0)
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/apt update
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1373]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1373]: pam_unix(cron:session): session closed for user root
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/apt install fail2ban -y
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/nano /etc/fail2ban/jail.local
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1683]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1683]: pam_unix(cron:session): session closed for user root
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/systemctl restart fail2ban
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
2026-04-03T02:15:31.752348+00:00 ubuntu: sshd[1711]: error: maximum authentication attempts exceeded for evenort12 from 192.168.56.1 port 56879 ssh2 [preauth]
2026-04-03T02:15:31.752348+00:00 ubuntu: sshd[1711]: Disconnecting authenticating user evenort12 192.168.56.1 port 56879: Too many authentication failures [preauth]
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1713]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1713]: pam_unix(cron:session): session closed for user root
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1781]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1781]: pam_unix(cron:session): session closed for user root
2026-04-03T02:15:31.752348+00:00 ubuntu: sshd[1896]: error: maximum authentication attempts exceeded for evenort12 from 192.168.56.1 port 57884 ssh2 [preauth]
2026-04-03T02:15:31.752348+00:00 ubuntu: sshd[1896]: Disconnecting authenticating user evenort12 192.168.56.1 port 57884: Too many authentication failures [preauth]
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1905]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-03T02:15:31.752348+00:00 ubuntu: CRON[1905]: pam_unix(cron:session): session closed for user root
2026-04-03T02:15:31.752348+00:00 ubuntu: sshd[1908]: Accepted password for evenort12 from 192.168.56.1 port 61330 ssh2
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: evenort12 : TTY=pts/0 ; PWD=/home/evenort12 ; USER=root ; COMMAND=/usr/bin/fail2ban-client status
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session opened for user root(uid=0) by evenort12(uid=1000)
2026-04-03T02:15:31.752348+00:00 ubuntu: sudo: pam_unix(sudo:session): session closed for user root
evenort12@ubuntu:~$
```

Figure 8, 9, 10, 11 : Ces images attestent que le service Fail2ban fonctionne correctement en utilisant fail2ban-client, puis en affichant les logs SSH pour vérifier les tentatives de connexion (voir pages 7 à 9).

```
evenort12@ubuntu:~$ grep "Failed password" /var/log/auth.log
2026-04-01T23:45:46.491038+00:00 ubuntu: sshd[4835]: Failed password for invalid user evenort12 from 127.0.0.1 port 38684 ssh2
2026-04-01T23:47:12.646075+00:00 ubuntu: sshd[4838]: Failed password for invalid user admin from 127.0.0.1 port 43780 ssh2
2026-04-01T23:47:27.131844+00:00 ubuntu: sshd[4838]: message repeated 2 times: [ Failed password for invalid user admin from 127.0.0.1 port 43780 ssh2]
evenort12@ubuntu:~$
```

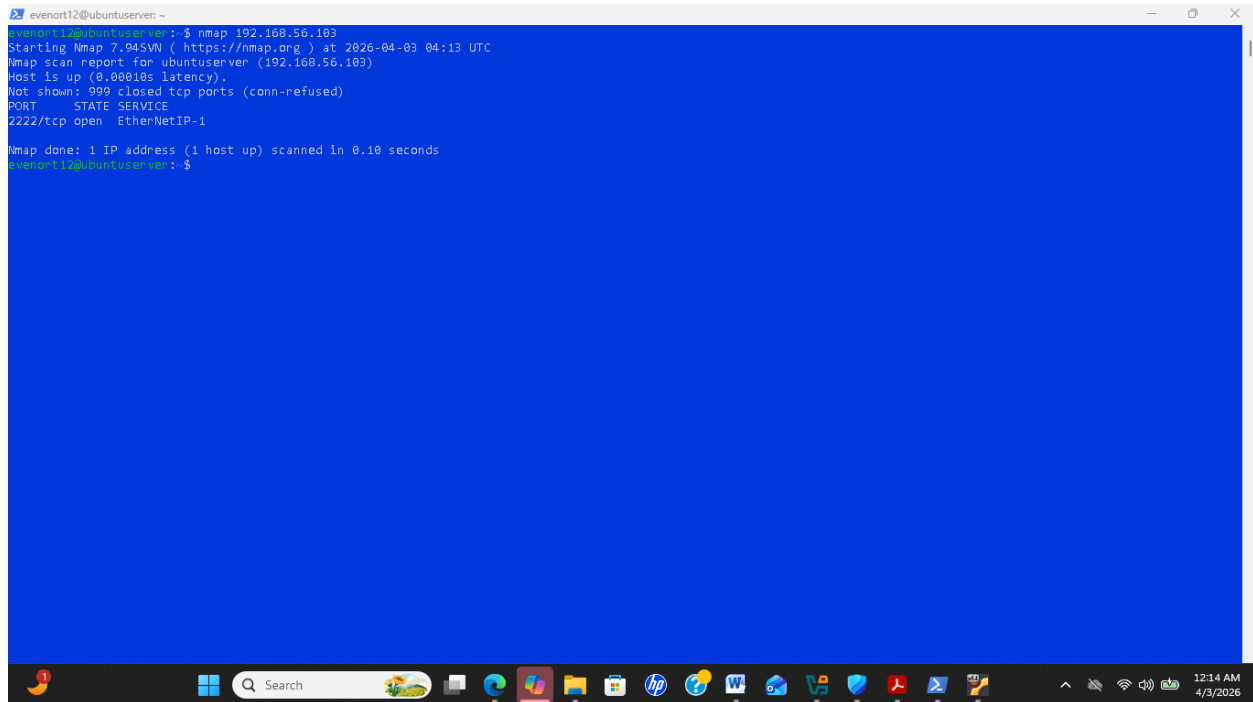
Figure 12 : Cette image atteste que la filtration des erreurs a été réalisée avec la commande **grep "Failed password" /var/log/auth.log**, afin d'identifier les tentatives de connexion SSH échouées.

```
evenort12@ubuntu:~$ cat /var/log/fail2ban.log
2026-04-03 03:06:58,720 fail2ban.server [1608]: INFO Starting Fail2ban v1.0.2
2026-04-03 03:06:58,723 fail2ban.observer [1608]: INFO Observer start...
2026-04-03 03:06:58,729 fail2ban.database [1608]: INFO Connected to fail2ban persistent database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-04-03 03:06:58,735 fail2ban.database [1608]: WARNING New database created. Version '4'
2026-04-03 03:06:58,736 fail2ban.jail [1608]: INFO Creating new jail 'sshd'
2026-04-03 03:06:59,603 fail2ban.jail [1608]: INFO Jail 'sshd' uses systemd {}
2026-04-03 03:06:59,604 fail2ban.jail [1608]: INFO Initiated 'systemd' backend
2026-04-03 03:06:59,605 fail2ban.filter [1608]: INFO maxLines: 1
2026-04-03 03:06:59,621 fail2ban.filtersystemd [1608]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.service + _COMM=sshd'
2026-04-03 03:06:59,621 fail2ban.filter [1608]: INFO maxRetry: 5
2026-04-03 03:06:59,622 fail2ban.filter [1608]: INFO findTime: 600
2026-04-03 03:06:59,622 fail2ban.actions [1608]: INFO banTime: 600
2026-04-03 03:06:59,622 fail2ban.filter [1608]: INFO encoding: UTF-8
2026-04-03 03:06:59,625 fail2ban.jail [1608]: INFO Jail 'sshd' started
2026-04-03 03:06:59,689 fail2ban.filtersystemd [1608]: INFO [sshd] Jail is in operation now (process new journal entries)
2026-04-03 03:13:19,579 fail2ban.server [1608]: INFO Shutdown in progress...
2026-04-03 03:13:19,579 fail2ban.observer [1608]: INFO Observer stop ... try to end queue 5 seconds
2026-04-03 03:13:19,600 fail2ban.observer [1608]: INFO Observer stopped, 0 events remaining.
2026-04-03 03:13:19,644 fail2ban.server [1608]: INFO Stopping all jails
2026-04-03 03:13:19,704 fail2ban.actions [1608]: NOTICE [sshd] Flush ticket(s) with nftables
2026-04-03 03:13:19,802 fail2ban.jail [1608]: INFO Jail 'sshd' stopped
2026-04-03 03:13:19,802 fail2ban.database [1608]: INFO Connection to database closed.
2026-04-03 03:13:19,802 fail2ban.server [1608]: INFO Exiting Fail2ban
2026-04-03 03:13:20,014 fail2ban.server [1701]: INFO Starting Fail2ban v1.0.2
2026-04-03 03:13:20,016 fail2ban.observer [1701]: INFO Observer start...
2026-04-03 03:13:20,022 fail2ban.database [1701]: INFO Connected to fail2ban persistent database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-04-03 03:13:20,024 fail2ban.jail [1701]: INFO Creating new jail 'sshd'
2026-04-03 03:13:20,146 fail2ban.jail [1701]: INFO Jail 'sshd' uses systemd {}
2026-04-03 03:13:20,148 fail2ban.jail [1701]: INFO Initiated 'systemd' backend
2026-04-03 03:13:20,149 fail2ban.filter [1701]: INFO maxLines: 1
2026-04-03 03:13:20,163 fail2ban.filtersystemd [1701]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.service + _COMM=sshd'
2026-04-03 03:13:20,163 fail2ban.filter [1701]: INFO maxRetry: 3
2026-04-03 03:13:20,163 fail2ban.filter [1701]: INFO findTime: 600
2026-04-03 03:13:20,163 fail2ban.actions [1701]: INFO banTime: 600
2026-04-03 03:13:20,163 fail2ban.filter [1701]: INFO encoding: UTF-8
2026-04-03 03:13:20,167 fail2ban.jail [1701]: INFO Jail 'sshd' started
2026-04-03 03:13:20,167 fail2ban.filtersystemd [1701]: INFO [sshd] Jail is in operation now (process new journal entries)
2026-04-03 03:14:44,684 fail2ban.filter [1701]: INFO [sshd] Found 192.168.56.1 - 2026-04-03 03:14:44
2026-04-03 03:22:45,919 fail2ban.filter [1701]: INFO [sshd] Found 192.168.56.1 - 2026-04-03 03:22:45
```

Figure 13 : Sur cette image, j'ai affiché les logs de Fail2ban afin de voir les processus en cours.

6 Test de sécurité

Scanner les ports du serveur avec : nmap 192.168.56.103



```
evenort12@ubuntu:~$ nmap 192.168.56.103
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-03 04:13 UTC
Nmap scan report for ubuntu:192.168.56.103
Host is up (0.00010s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
2222/tcp  open  EtherNetIP-1
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds
evenort12@ubuntu:~$
```

Figure 14 : Sur cette image, j'ai exécuté la commande `nmap <ip-server>` afin de scanner les ports ouverts du serveur et vérifier leur accessibilité.

Audit du serveur

7 Installation de Lynis

7.1 Installer l'outil :

`sudo apt update`

`sudo apt install lynis -y`

7.2 Vérifier l'installation :

`lynis -version`

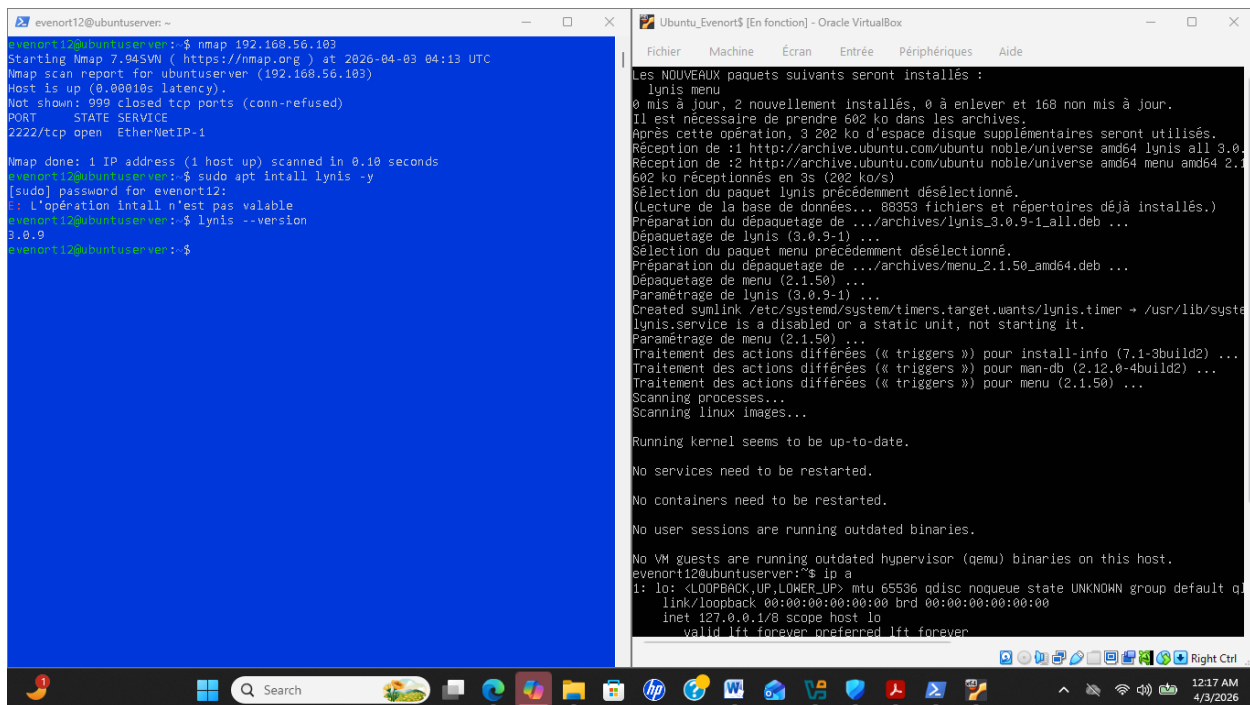
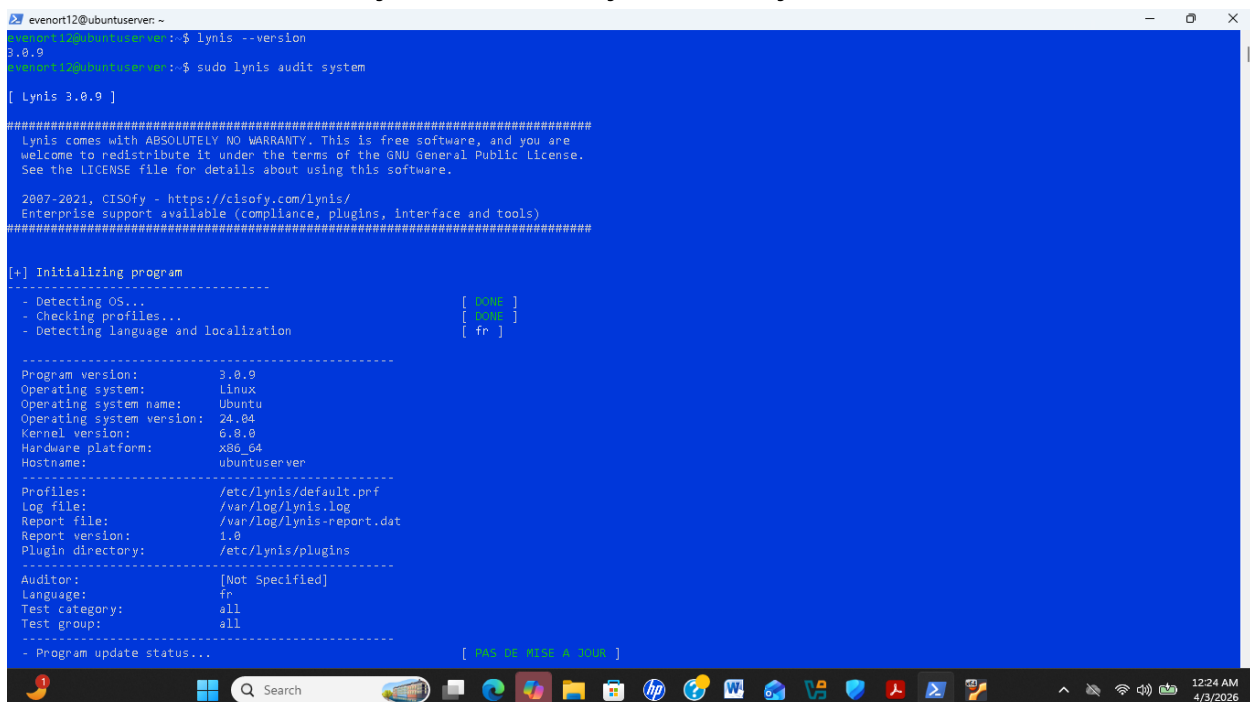


Figure 15: Cette image atteste l'installation de l'outil **Lynis** ainsi que la version utilisée pour l'audit de sécurité du serveur.

8 Lancer un audit complet

8.1 Exécuter un audit système : sudo lynis audit system




```
evenort12@ubuntu:~$ sudo dpkg-reconfigure --frontend=dialog base-system
- systemd-udev.service: [ MOYEN ]
- thermald.service: [ RISQUE ]
- tpm-udev.service: [ RISQUE ]
- ubuntu-advantage.service: [ RISQUE ]
- udisks2.service: [ RISQUE ]
- unattended-upgrades.service: [ RISQUE ]
- user@1000.service: [ RISQUE ]
- uuldd.service: [ MOYEN ]
- vgauth.service: [ RISQUE ]

[+] Noyau
- Checking default run level [ NON TROUVÉ ]
- Checking CPU support (NX/PAE) [ Trouvé ]
  CPU support: PAE and/or NoeXecute supported
- Checking kernel version and release [ FAIT ]
- Checking kernel type [ FAIT ]
- Checking loaded kernel modules [ FAIT ]
  Found 87 active modules
- Checking Linux kernel configuration file [ Trouvé ]
- Checking default I/O kernel scheduler [ NON TROUVÉ ]
- Checking for available kernel update [ OK ]
- Checking core dumps configuration
  configuration in systemd.conf files [ PAR DÉFAUT ]
  configuration in /etc/profile [ PAR DÉFAUT ]
  'hard' configuration in /etc/security/limits.conf [ PAR DÉFAUT ]
  'soft' configuration in /etc/security/limits.conf [ PAR DÉFAUT ]
- Checking retpoline core dumps configuration [ PROTÉGÉ ]
- Check if reboot is needed [ NON ]

[+] Mémoire et processus
- Checking /proc/meminfo [ Trouvé ]
- Searching for dead/zombie processes [ NON TROUVÉ ]
- Searching for IO waiting processes [ NON TROUVÉ ]
- Search prelink tooling [ NON TROUVÉ ]

[+] Utilisateurs, groupes et authentification
- Administrator accounts [ OK ]
- Unique UIDs [ OK ]
- Consistency of group files (grpck) [ OK ]
- Unique group IDs [ OK ]

[+] Sécurité
- LDAP module in PAM [ NON TROUVÉ ]
- Accounts without expire date [ SUGGESTION ]
- Accounts without password [ OK ]
- Locked accounts [ OK ]
- Checking user password aging (minimum) [ DÉSACTIVÉ ]
- User password aging (maximum) [ DÉSACTIVÉ ]
- Checking expired passwords [ OK ]
- Checking Linux single user mode authentication [ OK ]
- Determining default umask
  umask (/etc/profile) [ NON TROUVÉ ]
  umask (/etc/login.defs) [ SUGGESTION ]
- LDAP authentication support [ NON ACTIVE ]
- Logging failed login attempts [ ACTIVE ]

[+] Shells
- Checking shells from /etc/shells
  Result: found 9 shells (valid shells: 9).
- Session timeout settings/tools [ AUCUN ]
- Checking default umask values
  Checking default umask in /etc/bash.bashrc [ AUCUN ]
  Checking default umask in /etc/profile [ AUCUN ]

[+] Systèmes de fichier
- Checking mount points
  Checking /home mount point [ SUGGESTION ]
  Checking /tmp mount point [ SUGGESTION ]
  Checking /var mount point [ SUGGESTION ]
- Checking LVM volume groups [ Trouvé ]
- Checking LVM volumes [ Trouvé ]
- Query swap partitions (fstab) [ OK ]
- Testing swap partitions [ OK ]
- Testing /proc mount (hidepid) [ SUGGESTION ]
- Checking for old files in /tmp [ OK ]
- Checking /tmp sticky bit [ OK ]
- Checking /var/tmp sticky bit [ OK ]
- ACL support root file system [ ACTIVE ]
- Mount options of / [ OK ]
- Mount options of /boot [ PAR DÉFAUT ]
- Mount options of /dev [ PARTIELLEMENT RENFORCÉ ]
- Mount options of /dev/shm [ PARTIELLEMENT RENFORCÉ ]
- Mount options of /run [ RENFORCÉ ]
```



```
evenort12@ubuntu:~$ sudo dpkg --get-selections | grep -E "(install|hold)" | sed 's/hold/keep/' | xargs dpkg --get-selections | xargs dpkg --reconfigure
[+] Périphériques USB
-----
- Checking usb-storage driver (modprobe config) [ NON DESACTIVÉ ]
- Checking USB devices authorization [ ACTIVÉ ]
- Checking USBGuard [ NON TROUVÉ ]

[+] Stockage
-----
- Checking firewire ohci driver (modprobe config) [ DESACTIVÉ ]

[+] NFS
-----
- Check running NFS daemon [ NON TROUVÉ ]

[+] Services de noms
-----
- Checking search domains [ TROUVÉ ]
- Checking /etc/resolv.conf options [ TROUVÉ ]
- Searching DNS domain name [ INCONNU ]
- Checking /etc/hosts [ ACTIVÉ ]
- Duplicate entries in hosts file [ NON TROUVÉ ]
- Presence of configured hostname in /etc/hosts [ NON TROUVÉ ]
- Hostname mapped to localhost [ NON TROUVÉ ]
- Localhost mapping to IP address [ OK ]

[+] Ports et packages
-----
- Searching package managers [ TROUVÉ ]
- Searching dpkg package manager [ TROUVÉ ]
- Querying package manager [ TROUVÉ ]

[WARNING]: Test PKGS-7345 had a long execution: 12.201743 seconds

- Query unpurged packages [ ACTIVÉ ]
- Checking security repository in sources.list.d directory [ OK ]
- Checking APT package database [ OK ]

W: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble/InRelease Temporary failure resolving 'archive.ubuntu.com'
W: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble-updates/InRelease Temporary failure resolving 'archive.ubuntu.com'
W: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble-backports/InRelease Temporary failure resolving 'archive.ubuntu.com'
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/noble-security/InRelease Temporary failure resolving 'security.ubuntu.com'
W: Some index files failed to download. They have been ignored, or old ones used instead.
- Checking vulnerable packages [ AVERTISSEMENT ]

[+] Mise en réseau
-----
- Checking IPv6 configuration [ ACTIVÉ ]
- Configuration method [ AUTO ]
- IPv6 only [ NON ]
- Checking configured nameservers [ TROUVÉ ]
- Testing nameservers [ OK ]
- Nameserver: 127.0.0.53 [ OK ]
- DNSSEC supported (systemd-resolved) [ INCONNU ]
- Getting listening ports (TCP/UDP) [ ÉCOUTÉ ]
- Checking promiscuous interfaces [ OK ]
- Checking status DHCP client [ NON TROUVÉ ]
- Checking for ARP monitoring software [ NON TROUVÉ ]
- Uncommon network protocols [ 0 ]

[+] Imprimantes et serveurs d'impression
-----
- Checking cups daemon [ NON TROUVÉ ]
- Checking lp daemon [ NON LANCÉ ]

[+] Logiciel : Email et messagerie
-----

[+] Logiciel : Pare-feu
-----
- Checking iptables kernel module [ TROUVÉ ]
- Checking iptables policies of chains [ TROUVÉ ]
- Checking for empty ruleset [ OK ]
- Checking for unused rules [ TROUVÉ ]
- Checking host based firewall [ ACTIVÉ ]

[+] Logiciel : Serveur web
-----
- Checking Apache [ NON TROUVÉ ]
```



```
evenort12@ubuntu:~$
- Checking Apache [ NON TROUVÉ ]
- Checking nginx [ NON TROUVÉ ]

[+] Prise en charge SSH
-----
- Checking running SSH daemon [ Trouvé ]
- Searching SSH configuration [ Trouvé ]
- OpenSSH option: AllowTcpForwarding [ SUGGESTION ]
- OpenSSH option: ClientAliveCountMax [ SUGGESTION ]
- OpenSSH option: ClientAliveInterval [ OK ]
- OpenSSH option: FingerPrintHash [ OK ]
- OpenSSH option: GatewayPorts [ OK ]
- OpenSSH option: IgnoreRhosts [ OK ]
- OpenSSH option: LoginGraceTime [ OK ]
- OpenSSH option: LogLevel [ SUGGESTION ]
- OpenSSH option: MaxAuthTries [ OK ]
- OpenSSH option: MaxSessions [ SUGGESTION ]
- OpenSSH option: PermitRootLogin [ OK ]
- OpenSSH option: PermitUserEnvironment [ OK ]
- OpenSSH option: PermitTunnel [ OK ]
- OpenSSH option: Port [ OK ]
- OpenSSH option: PrintLastLog [ OK ]
- OpenSSH option: StrictModes [ OK ]
- OpenSSH option: TCPKeepAlive [ SUGGESTION ]
- OpenSSH option: UseDNS [ OK ]
- OpenSSH option: X11Forwarding [ OK ]
- OpenSSH option: AllowAgentForwarding [ SUGGESTION ]
- OpenSSH option: AllowUsers [ Trouvé ]
- OpenSSH option: AllowGroups [ NON TROUVÉ ]

[+] Prise en charge SNMP
-----
- Checking running SNMP daemon [ NON TROUVÉ ]

[+] Bases de données
-----
No database engines found

[+] Services LDAP
-----
- Checking OpenLDAP instance [ NON TROUVÉ ]

[+] PHP
-----
- Checking PHP [ NON TROUVÉ ]

[+] Prise en charge Squid
-----
- Checking running Squid daemon [ NON TROUVÉ ]

[+] Journalisation et fichiers
-----
- Checking for a running log daemon [ OK ]
- Checking Syslog-NG status [ NON TROUVÉ ]
- Checking systemd journal status [ OK ]
- Checking Metalog status [ NON TROUVÉ ]
- Checking Rsyslog status [ Trouvé ]
- Checking RFC 3195 daemon status [ NON TROUVÉ ]
- Checking minilogd instances [ NON TROUVÉ ]
- Checking logrotate presence [ OK ]
- Checking remote logging [ NON ACTIVÉ ]
- Checking log directories (static list) [ FAIBLE ]
- Checking open log files [ FAIBLE ]
- Checking deleted files in use [ FAIBLE ]

[+] Services non sécurisés
-----
- Installed inetd package [ NON TROUVÉ ]
- Installed xinetd package [ OK ]
- xinetd status [ OK ]
- Installed rsh client package [ OK ]
- Installed rsh server package [ OK ]
- Installed telnet client package [ OK ]
- Installed telnet server package [ NON TROUVÉ ]
- Checking NIS client installation [ OK ]
- Checking NIS server installation [ OK ]
- Checking TFTP client installation [ OK ]
- Checking TFTP server installation [ OK ]

[+] Bannières et identification
-----
- /etc/issue [ Trouvé ]
- /etc/issue contents [ FAIBLE ]
- /etc/issue.net [ Trouvé ]
- /etc/issue.net contents [ FAIBLE ]
```

```
evenort12@ubuntu:server ~  
[WARNINGS]: Test CRYP-7902 had a long execution: 34.332530 seconds  
- Found 0 encrypted and 1 unencrypted swap devices in use. [ OK ]  
- Kernel entropy is sufficient [ OK ]  
- HW RNG & rngd [ NON ]  
- SW prng [ NON ]  
- MOR variable not found [ FAIBLE ]  
[+] Virtualisation  
-----  
[+] Conteneurs  
-----  
[+] Frameworks de sécurité  
-----  
- Checking presence AppArmor [ Trouvé ]  
- Checking AppArmor status [ Activé ]  
- Found 33 unconfinned processes [ NON TROUVÉ ]  
- Checking presence SELinux [ NON TROUVÉ ]  
- Checking presence TOMOYO Linux [ NON TROUVÉ ]  
- Checking presence grsecurity [ NON TROUVÉ ]  
- Checking for implemented MAC framework [ OK ]  
[+] Logiciel : Intégrité de fichier  
-----  
- Checking file integrity tools [ DÉSACTIVÉ ]  
- dm-integrity (status) [ DÉSACTIVÉ ]  
- Checking presence integrity tool [ NON TROUVÉ ]  
[+] Logiciel : System tooling  
-----  
- Checking automation tooling [ NON TROUVÉ ]  
- Automation tooling [ Trouvé ]  
- Checking presence of Fail2ban [ Trouvé ]  
2026-04-03 04:27:32,705 fail2ban.configreader [35582]: WARNING 'allowip6' not defined in 'Definition'. Using default one: 'auto'  
- Checking Fail2ban jails [ Activé ]  
- Checking for IDS/IPS tooling [ Trouvé ]  
[+] Logiciel : Malveillants  
-----  
pgrep: pattern that searches for process name longer than 15 characters will result in zero matches  
Try 'pgrep -f' option to match against the complete command line.  
- Malware software components [ NON TROUVÉ ]  
[+] Permissions de fichier  
-----  
- Starting file permissions check  
File: /boot/grub/grub.cfg [ OK ]  
File: /etc/crontab [ SUGGESTION ]  
File: /etc/group [ OK ]  
File: /etc/group- [ OK ]  
File: /etc/hosts.allow [ OK ]  
File: /etc/hosts.deny [ OK ]  
File: /etc/issue [ OK ]  
File: /etc/issue.net [ OK ]  
File: /etc/passwd [ OK ]  
File: /etc/passwd- [ OK ]  
File: /etc/ssh/ssh_config [ SUGGESTION ]  
Directory: /root/.ssh [ OK ]  
Directory: /etc/cron.d [ SUGGESTION ]  
Directory: /etc/cron.daily [ SUGGESTION ]  
Directory: /etc/cron.hourly [ SUGGESTION ]  
Directory: /etc/cron.weekly [ SUGGESTION ]  
Directory: /etc/cron.monthly [ SUGGESTION ]  
[+] Dossiers personnels  
-----  
- Permissions of home directories [ OK ]  
- Ownership of home directories [ OK ]  
- Checking shell history files [ OK ]  
[+] Kernel Hardening  
-----  
- Comparing sysctl key pairs with scan profile  
- dev.tty.ldisc_autoload (exp: 0) [ DIFFÉRENT ]  
- fs.protected_files (exp: 2) [ DIFFÉRENT ]  
- fs.protected_hardlinks (exp: 1) [ OK ]  
- fs.protected_regular (exp: 2) [ OK ]  
- fs.protected_symlinks (exp: 1) [ OK ]  
- fs.suid_dumpable (exp: 0) [ DIFFÉRENT ]  
- kernel.core_uses_pid (exp: 1) [ DIFFÉRENT ]  
- kernel.ctrl-alt-del (exp: 0) [ OK ]
```

```
evenort12@ubuntu:server ~  
- net.ipv4.conf.all.bootp_relay (exp: 0) [ OK ]  
- net.ipv4.conf.all.forwarding (exp: 0) [ OK ]  
- net.ipv4.conf.all.log_martians (exp: 1) [ DIFFÉRENT ]  
- net.ipv4.conf.all.mc_forwarding (exp: 0) [ OK ]  
- net.ipv4.conf.all.proxy_arp (exp: 0) [ OK ]  
- net.ipv4.conf.all.rp_filter (exp: 1) [ DIFFÉRENT ]  
- net.ipv4.conf.all.send_redirects (exp: 0) [ DIFFÉRENT ]  
- net.ipv4.conf.default.accept_redirects (exp: 0) [ OK ]  
- net.ipv4.conf.default.accept_source_route (exp: 0) [ DIFFÉRENT ]  
- net.ipv4.conf.default.log_martians (exp: 1) [ DIFFÉRENT ]  
- net.ipv4.icmp_echo_ignore_broadcasts (exp: 1) [ OK ]  
- net.ipv4.icmp_ignore_bogus_error_responses (exp: 1) [ OK ]  
- net.ipv4.tcp_syncookies (exp: 1) [ OK ]  
- net.ipv4.tcp_timestamps (exp: 0 1) [ OK ]  
- net.ipv6.conf.all.accept_redirects (exp: 0) [ OK ]  
- net.ipv6.conf.all.accept_source_route (exp: 0) [ OK ]  
- net.ipv6.conf.default.accept_redirects (exp: 0) [ OK ]  
- net.ipv6.conf.default.accept_source_route (exp: 0) [ OK ]  
  
[+] Hardening  
-----  
- Installed compiler(s) [ NON TROUVÉ ]  
- Installed malware scanner [ NON TROUVÉ ]  
- Non-native binary formats [ TROUVÉ ]  
  
[+] Tests personnalisés  
-----  
- Running custom tests... [ AUCUN ]  
  
[+] Plugins (phase 2)  
-----  
  
-----  
-[ Lynis 3.0.9 Results ]-  
  
Warnings (1):  
-----  
| Found one or more vulnerable packages. [PKGS-7392]  
| https://cisofy.com/lynis/controls/PKGS-7392/  
  
Suggestions (45):  
-----  
  
https://cisofy.com/lynis/controls/LOGG-2154/  
  
* Add a legal banner to /etc/issue, to warn unauthorized users [BANN-7126]  
| https://cisofy.com/lynis/controls/BANN-7126/  
  
* Add legal banner to /etc/issue.net, to warn unauthorized users [BANN-7130]  
| https://cisofy.com/lynis/controls/BANN-7130/  
  
* Enable process accounting [ACCT-9622]  
| https://cisofy.com/lynis/controls/ACCT-9622/  
  
* Enable sysstat to collect accounting (disabled) [ACCT-9626]  
| https://cisofy.com/lynis/controls/ACCT-9626/  
  
* Enable auditd to collect audit information [ACCT-9628]  
| https://cisofy.com/lynis/controls/ACCT-9628/  
  
* Install a file integrity tool to monitor changes to critical and sensitive files [FINT-4350]  
| https://cisofy.com/lynis/controls/FINT-4350/  
  
* Determine if automation tools are present for system management [TOOL-5002]  
| https://cisofy.com/lynis/controls/TOOL-5002/  
  
* Consider restricting file permissions [FILE-7524]  
- Details : see screen output or log file  
- Solution : Use chmod to change file permissions  
| https://cisofy.com/lynis/controls/FILE-7524/  
  
* One or more sysctl values differ from the scan profile and could be tweaked [KRNL-6000]  
- Solution : Change sysctl value or disable test (skip-test=KRNL-6000:<sysctl-key>)  
| https://cisofy.com/lynis/controls/KRNL-6000/  
  
* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]  
- Solution : Install a tool like rkhunter, chkrootkit, OSSEC  
| https://cisofy.com/lynis/controls/HRDN-7230/  
  
Follow-up:  
-----  
- Show details of a test (lynis show details TEST-ID)  
- Check the logfile for all details (less /var/log/lynis.log)  
- Read security controls texts (https://cisofy.com)  
- Use --upload to upload data to central system (Lynis Enterprise users)
```

```
evenort12@ubuntu:~$ sudo lynis --audit --report /var/log/lynis-report.dat
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

=====

Lynis security scan details:

Hardening index : 63 [##### ]
Tests performed : 257
Plugins enabled : 1

Components:
- Firewall [ ]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [ ]
- Vulnerability scan [ ]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOFY - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all settings)

evenort12@ubuntu:~$
```

Figure 16, a 29 : Ces images illustrent l'exécution d'un audit complet du système avec Lynis, afin d'obtenir le score initial de sécurité du serveur (voir pages 12 à 18).

9 Analyse du rapport

9.1 Les résultats sont enregistrés dans :

/var/log/lynis.log et /var/log/lynis-report.dat

9.2 Afficher les suggestions :

grep suggestion /var/log/lynis-report.dat

```
evenort12@ubuntu:~$ cat /var/log/lynis.log et /var/log/lynis-report.dat
-bash: /var/log/lynis.log: No such file or directory
evenort12@ubuntu:~$ cat /var/log/lynis-report.dat
-bash: /var/log/lynis-report.dat: Permission denied
evenort12@ubuntu:~$ sudo cat /var/log/lynis.log
cat: /var/log/lynis.log: No such file or directory
evenort12@ubuntu:~$ sudo cat /var/log/lynis-report.dat
cat: /var/log/lynis-report.dat: Permission denied
evenort12@ubuntu:~$ grep suggestion /var/log/lynis-report.dat
```

Figure 30: Sur cette image, j'ai tenté d'accéder aux fichiers de logs de Lynis avec /var/log/lynis.log et /var/log/lynis-report.dat, mais ils n'ont pas pu être trouvés sur le système.

```
evenort12@ubuntu:~$ sudo grep suggestion /var/log/lynis-report.dat
suggestion[ ]=LYNIS-0000|This release is more than 4 months old. Check the website or GitHub to see if there is an update available.|-|-|
suggestion[ ]=DEB-0200|Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions.|-|-|
suggestion[ ]=DEB-0810|Install apt-listbugs to display a list of critical bugs prior to each APT installation.|-|-|
suggestion[ ]=DEB-0811|Install apt-listchanges to display any significant changes prior to any upgrade via APT.|-|-|
suggestion[ ]=BOOT-5122|Set a password on GRUB boot loader to prevent altering boot configuration (e.g. boot in single user mode without password).|-|-|
suggestion[ ]=BOOT-5264|Consider hardening system services|Run '/usr/bin/systemd-analyze security SERVICE' for each service|-|-|
suggestion[ ]=KRNL-5820|If not required, consider explicit disabling of core dump in /etc/security/limits.conf file.|-|-|
suggestion[ ]=AUTH-9220|Check PAM configuration, add rounds if applicable and expire passwords to encrypt with new values.|-|-|
suggestion[ ]=AUTH-9230|Configure password hashing rounds in /etc/login.defs.|-|-|
suggestion[ ]=AUTH-9262|Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc.|-|-|
suggestion[ ]=AUTH-9282|When possible set expire dates for all password protected accounts.|-|-|
suggestion[ ]=AUTH-9286|Configure minimum password age in /etc/login.defs.|-|-|
suggestion[ ]=AUTH-9286|Configure maximum password age in /etc/login.defs.|-|-|
suggestion[ ]=AUTH-9328|Default umask in /etc/login.defs could be more strict like 027.|-|-|
suggestion[ ]=FILE-6310|To decrease the impact of a full /home file system, place /home on a separate partition.|-|-|
suggestion[ ]=FILE-6310|To decrease the impact of a full /tmp file system, place /tmp on a separate partition.|-|-|
suggestion[ ]=FILE-6310|To decrease the impact of a full /var file system, place /var on a separate partition.|-|-|
suggestion[ ]=USB-1000|Disable drivers like USB storage when not used, to prevent unauthorized storage or data theft.|-|-|
suggestion[ ]=NAME-4628|Check DNS configuration for the dns domain name.|-|-|
suggestion[ ]=NAME-4604|Add the IP name and FQDN to /etc/hosts for proper name resolving.|-|-|
suggestion[ ]=PKGS-7378|Install debsums utility for the verification of packages with known good database.|-|-|
suggestion[ ]=PKGS-7392|Update your system with apt-get update, apt-get upgrade, apt-get dist-upgrade and/or unattended-upgrades.|-|-|
suggestion[ ]=PKGS-7394|Install package apt-show-versions for patch management purposes.|-|-|
suggestion[ ]=NETW-3200|Determine if protocol 'dccp' is really needed on this system.|-|-|
suggestion[ ]=NETW-3200|Determine if protocol 'sctp' is really needed on this system.|-|-|
suggestion[ ]=NETW-3200|Determine if protocol 'rds' is really needed on this system.|-|-|
suggestion[ ]=NETW-3200|Determine if protocol 'tipc' is really needed on this system.|-|-|
suggestion[ ]=FIRE-4513|Check iptables rules to see which rules are currently not used.|-|-|
suggestion[ ]=SSH-7408|Consider hardening SSH configuration|AllowTcpForwarding (set YES to NO).|-|-|
suggestion[ ]=SSH-7408|Consider hardening SSH configuration|ClientAliveCountMax (set 3 to 2).|-|-|
suggestion[ ]=SSH-7408|Consider hardening SSH configuration|LogLevel (set INFO to VERBOSE).|-|-|
suggestion[ ]=SSH-7408|Consider hardening SSH configuration|MaxSessions (set 10 to 2).|-|-|
suggestion[ ]=SSH-7408|Consider hardening SSH configuration|TCPKeepAlive (set YES to NO).|-|-|
suggestion[ ]=SSH-7408|Consider hardening SSH configuration|AllowAgentForwarding (set YES to NO).|-|-|
suggestion[ ]=LOGG-2154|Enable logging to an external logging host for archiving purposes and additional protection.|-|-|
suggestion[ ]=BANN-7126|Add a legal banner to /etc/issue, to warn unauthorized users.|-|-|
suggestion[ ]=BANN-7130|Add legal banner to /etc/issue.net, to warn unauthorized users.|-|-|
suggestion[ ]=ACCT-9622|Enable process accounting.|-|-|
suggestion[ ]=ACCT-9626|Enable sysstat to collect accounting (disabled).|-|-|
suggestion[ ]=ACCT-9628|Enable auditd to collect audit information.|-|-|
suggestion[ ]=FINT-4350|Install a file integrity tool to monitor changes to critical and sensitive files.|-|-|
suggestion[ ]=TOOL-5802|Determine if automation tools are present for system management.|-|-|
```

Figure 31: Cette image illustre l’affichage des différentes suggestions fournies par Lynis à l’issue de l’audit de sécurité.

10. Amélioration de la sécurité

Identifier 3 recommandations Lynis et les appliquer.

Exemples possibles :

10.1 Installer un firewall

sudo apt install ufw

10.2 Installer un IDS

sudo apt install aide

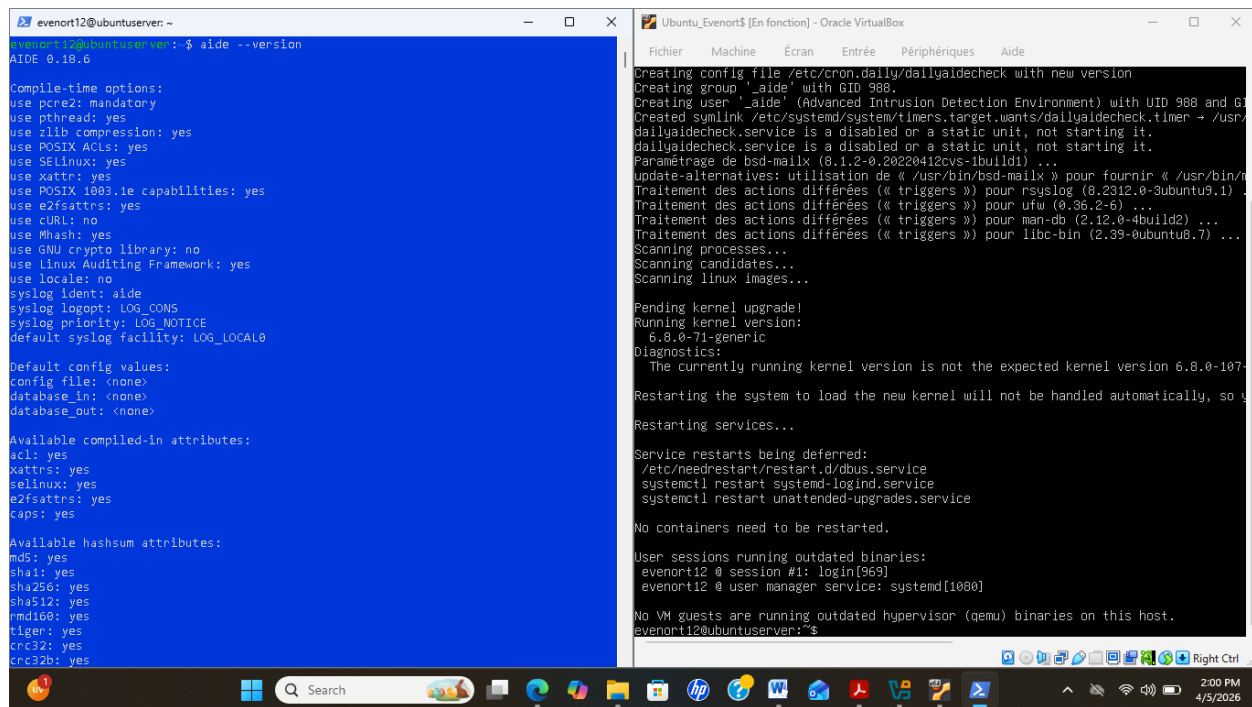


Figure 32: Cette image illustre l'installation et version de l'outil vérifiées.

11.3 Outil:

Activer les mises à jour automatiques

`sudo apt install unattended-upgrades`

Relancer l'audit

12. Après les corrections :

`sudo lynis audit system`

```
evenort12@ubuntu:~$ sudo lynis audit system
[sudo] password for evenort12:

[ Lynis 3.0.9 ]

#####
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2021, CISOFY - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####

[+] Initializing program
-----
- Detecting OS... [ FOUND ]
- Checking profiles... [ FOUND ]
- Detecting language and localization [ fr ]

-----
Program version: 3.0.9
Operating system: Linux
Operating system name: Ubuntu
Operating system version: 24.04
Kernel version: 6.8.0
Hardware platform: x86_64
Hostname: ubuntu

-----
Profiles: /etc/lynis/default.pnf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins

-----
Auditor: [Not Specified]
Language: fr
Test category: all
Test group: all

[+] Test group: all
-----
- Program update status... [ HAS DE WISE A JOIN ]

[+] Outils système
-----
- Scanning available tools...
- Checking system binaries...

[+] Plugins (phase 1)
-----
Note : Les plugins ont des tests plus poussés qui peuvent prendre plusieurs minutes

- Plugin: debian
|
[+] Debian Tests
-----
- Checking for system binaries that are required by Debian Tests...
- Checking /bin... [ FOUND ]
- Checking /sbin... [ FOUND ]
- Checking /usr/bin... [ FOUND ]
- Checking /usr/sbin... [ FOUND ]
- Checking /usr/local/bin... [ FOUND ]
- Checking /usr/local/sbin... [ FOUND ]
- Authentication:
- PAM (Pluggable Authentication Modules):

[WARNING]: Test DEB-0001 had a long execution: 62.868048 seconds

- libpam-tmpdir [ Not Installed ]
- File System Checks:
- DM-Crypt, Cryptsetup & Cryptmount:
- Checking / on /dev/sda3 [ NOT ENCRYPTED ]
- Checking /boot on /dev/sda2 [ NOT ENCRYPTED ]
- Software:
- apt-listbugs [ Not Installed ]
- apt-listchanges [ Not Installed ]
- needrestart [ Not Installed ]
- fail2ban [ Installation -> jail.local ]

[+] Démarrage et services
-----
```

```
evenort12@ubuntu:~$ sudo systemctl status --all
[+] Démarrage et services
-----
- Service Manager [ OK ]
- Checking UEFI boot [ DÉSACTIVÉ ]
- Checking presence GRUB2 [ TROUVÉ ]
- Checking for password protection [ AUCUN ]
- Check running services (systemctl) [ FAIT ]
  Result: found 20 running services
- Check enabled services at boot (systemctl) [ FAIT ]
  Result: found 51 enabled services
- Check startup files (permissions) [ OK ]
- Running 'systemd-analyze security'
  - ModemManager.service: [ MOYEN ]
  - cron.service: [ RISQUE ]
  - dbus.service: [ RISQUE ]
  - dm-event.service: [ RISQUE ]
  - dmesg.service: [ RISQUE ]
  - emergency.service: [ RISQUE ]
  - fail2ban.service: [ RISQUE ]
  - getty@tty1.service: [ RISQUE ]
  - iscsid.service: [ RISQUE ]
  - lvm2-lvmopld.service: [ RISQUE ]
  - lvm2-lvmetad.service: [ RISQUE ]
  - multipathd.service: [ RISQUE ]
  - networkd-dispatcher.service: [ RISQUE ]
  - open-vm-tools.service: [ RISQUE ]
  - plymouth-start.service: [ RISQUE ]
  - polkit.service: [ PROTÉGÉ ]
  - postfix@-.service: [ RISQUE ]
  - rc-local.service: [ RISQUE ]
  - rescue.service: [ RISQUE ]
  - rsyncd.service: [ MOYEN ]
  - snapd.service: [ RISQUE ]
  - ssh.service: [ RISQUE ]
  - systemd-ask-password-console.service: [ RISQUE ]
  - systemd-ask-password-plymouth.service: [ RISQUE ]
  - systemd-ask-password-wall.service: [ RISQUE ]
  - systemd-bsod.service: [ RISQUE ]
  - systemd-fsckd.service: [ RISQUE ]
  - systemd-initctl.service: [ RISQUE ]
  - systemd-journald.service: [ PROTÉGÉ ]

[+] Noyau
-----
- Checking default run level [ RUNLEVEL 1 ]
- Checking CPU support (NX/PAE) [ TROUVÉ ]
  CPU support: PAE and/or NoEXecute supported
- Checking kernel version and release [ FAIT ]
- Checking kernel type [ FAIT ]
- Checking loaded kernel modules [ FAIT ]
  Found 89 active modules
- Checking Linux kernel configuration file [ TROUVÉ ]
- Checking default I/O kernel scheduler [ NON TROUVÉ ]
- Checking for available kernel update [ OK ]
- Checking core dumps configuration
  - configuration in /etc/profile [ PAR DÉFAUT ]
  - 'hard' configuration in /etc/security/limits.conf [ PAR DÉFAUT ]
  - 'soft' configuration in /etc/security/limits.conf [ PAR DÉFAUT ]
- Checking setuid core dumps configuration [ PROTÉGÉ ]
- Check if reboot is needed [ OUI ]

[+] Mémoire et processus
-----
- Checking /proc/meminfo [ TROUVÉ ]
- Searching for dead/zombie processes [ NON TROUVÉ ]
- Searching for IO waiting processes [ NON TROUVÉ ]
- Search prelink tooling [ NON TROUVÉ ]

[+] Utilisateurs, groupes et authentification
-----
- Administrator accounts [ OK ]
- Unique UIDs [ OK ]
- Consistency of group files (grpck) [ OK ]
```



```
evenort12@ubuntu:~$ sudo searchprelinktooling
[ NON TROUVÉ ]

[+] Utilisateurs, groupes et authentification
-----
- Administrator accounts [ OK ]
- Unique UIDs [ OK ]
- Consistency of group files (grpck) [ OK ]
- Unique group IDs [ OK ]
- Unique group names [ OK ]
- Password file consistency [ OK ]
- Password hashing methods [ SUGGESTION ]
- Checking password hashing rounds [ DÉSACTIVÉ ]
- Query system users (non daemons) [ ÉLIGÉ ]
- NIS+ authentication support [ NON ACTIF ]
- NIS authentication support [ NON ACTIF ]
- Sudoers file(s) [ Trouvé ]
- Permissions for directory: /etc/sudoers.d [ AVERTISSEMENT ]
- Permissions for: /etc/sudoers [ OK ]
- Permissions for: /etc/sudoers.d/README [ OK ]
- PAM password strength tools [ SUGGESTION ]
- PAM configuration files (pam.conf) [ Trouvé ]
- PAM configuration files (pam.d) [ Trouvé ]
- PAM modules [ Trouvé ]
- LDAP module in PAM [ NON TROUVÉ ]
- Accounts without expire date [ SUGGESTION ]
- Accounts without password [ OK ]
- Locked accounts [ OK ]
- Checking user password aging (minimum) [ DÉSACTIVÉ ]
- User password aging (maximum) [ DÉSACTIVÉ ]
- Checking expired passwords [ OK ]
- Checking Linux single user mode authentication [ OK ]
- Determining default umask [ NON TROUVÉ ]
- umask (/etc/profile) [ SUGGESTION ]
- umask (/etc/login.defs) [ NON TROUVÉ ]
- LDAP authentication support [ NON ACTIF ]
- Logging failed login attempts [ ÉLIGÉ ]

[+] Shells
-----
- Checking shells from /etc/shells
  Result: found 9 shells (valid shells: 9). [ AUCUN ]
- Session timeout settings/tools
- Checking default umask values
```

```
evenort12@ubuntu:~$ sudo searchprelinktooling
[+] Systèmes de fichier
-----
- Checking mount points
  - Checking /home mount point [ SUGGESTION ]
  - Checking /tmp mount point [ SUGGESTION ]
  - Checking /var mount point [ SUGGESTION ]
- Checking LVM volume groups [ Trouvé ]
- Checking LVM volumes [ Trouvé ]
- Query swap partitions (fstab) [ OK ]
- Testing swap partitions [ OK ]
- Testing /proc mount (hidetpid) [ SUGGESTION ]
- Checking for old files in /tmp [ OK ]
- Checking /tmp sticky bit [ OK ]
- Checking /var/tmp sticky bit [ OK ]
- ACL support root file system [ ACTIF ]
- Mount options of / [ OK ]
- Mount options of /boot [ PAR DÉFAUT ]
- Mount options of /dev [ PARTIELLEMENT RENFORCÉ ]
- Mount options of /dev/shm [ PARTIELLEMENT RENFORCÉ ]
- Mount options of /run [ RENFORCÉ ]
- Total without nodev:5 noexec:7 nosuid:3 ro or noexec (W*X): 7 of total 22
- Disable kernel support of some filesystems

[+] Périphériques USB
-----
- Checking usb-storage driver (modprobe config) [ NON DESACTIVÉ ]
- Checking USB devices authorization [ ACTIF ]
- Checking USBGuard [ NON TROUVÉ ]

[+] Stockage
-----
- Checking firewire ohci driver (modprobe config) [ DÉSACTIVÉ ]

[+] NFS
-----
- Check running NFS daemon [ NON TROUVÉ ]

[+] Services de noms
-----
- Checking search domains [ Trouvé ]
- Checking /etc/resolv.conf options [ Trouvé ]
- Searching DNS domain name [ INCONNU ]
```

```
evenort12@ubuntuserver: ~  
[+] Ports et packages  
-----  
- Searching package managers  
- Searching dpkg package manager [ TROUVÉ ]  
- Querying package manager  
  
[WARNING]: Test PKGS-7345 had a long execution: 25.576034 seconds  
  
- Query unpurged packages [ AUCUN ]  
- Checking security repository in sources.list.d directory [ OK ]  
- Checking APT package database [ OK ]  
W: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble/InRelease Temporary failure resolving 'archive.ubuntu.com'  
W: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble-updates/InRelease Temporary failure resolving 'archive.ubuntu.com'  
W: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble-backports/InRelease Temporary failure resolving 'archive.ubuntu.com'  
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/noble-security/InRelease Temporary failure resolving 'security.ubuntu.com'  
W: Some index files failed to download. They have been ignored, or old ones used instead.  
- Checking vulnerable packages [ OK ]  
  
[WARNING]: Test PKGS-7392 had a long execution: 374.411282 seconds  
  
- Checking upgradeable packages [ IGNORÉ ]  
- Checking package audit tool [ INSTALLÉ ]  
Found: apt-check  
- Toolkit for automatic upgrades (unattended-upgrade) [ TROUVÉ ]  
  
[+] Mise en réseau  
-----  
- Checking IPv6 configuration [ ACTIVÉ ]  
Configuration method [ AUTO ]  
IPv6 only [ NON ]  
  
[WARNING]: Test NETW-2600 had a long execution: 11.814825 seconds  
  
- Checking configured nameservers  
- Testing nameservers  
Nameserver: 127.0.0.53 [ OK ]  
- DNSSEC supported (systemd-resolved) [ INCONNU ]  
- Getting listening ports (TCP/UDP) [ FAIT ]  
- Checking promiscuous interfaces [ OK ]  
- Checking status DHCP client  
- Checking for ARP monitoring software [ NON TROUVÉ ]  
- Uncommon network protocols [ 0 ]
```

```
evenort12@ubuntuserver: ~  
[+] Logiciel : Pare-feu  
-----  
- Checking iptables kernel module [ TROUVÉ ]  
- Checking iptables policies of chains [ TROUVÉ ]  
- Checking for empty ruleset [ OK ]  
- Checking for unused rules [ TROUVÉ ]  
- Checking host based firewall [ ACTIVÉ ]  
  
[+] Logiciel : Serveur web  
-----  
- Checking Apache [ NON TROUVÉ ]  
- Checking nginx [ NON TROUVÉ ]  
  
[+] Prise en charge SSH  
-----  
- Checking running SSH daemon [ TROUVÉ ]  
- Searching SSH configuration [ TROUVÉ ]  
- OpenSSH option: AllowTcpForwarding [ SUGGESTION ]  
- OpenSSH option: ClientAliveCountMax [ SUGGESTION ]  
- OpenSSH option: ClientAliveInterval [ OK ]  
- OpenSSH option: Fingerprinthash [ OK ]  
- OpenSSH option: GatewayPorts [ OK ]  
- OpenSSH option: IgnoreRhosts [ OK ]  
- OpenSSH option: LoginGraceTime [ OK ]  
- OpenSSH option: LogLevel [ SUGGESTION ]  
- OpenSSH option: MaxAuthTries [ OK ]  
- OpenSSH option: MaxSessions [ SUGGESTION ]  
- OpenSSH option: PermitRootLogin [ OK ]  
- OpenSSH option: PermitUserEnvironment [ OK ]  
- OpenSSH option: PermitTunnel [ OK ]  
- OpenSSH option: Port [ OK ]  
- OpenSSH option: PrintLastLog [ OK ]  
- OpenSSH option: StrictModes [ OK ]  
- OpenSSH option: TCPKeepAlive [ SUGGESTION ]  
- OpenSSH option: UseDNS [ OK ]  
- OpenSSH option: X11Forwarding [ OK ]  
- OpenSSH option: AllowAgentForwarding [ SUGGESTION ]  
- OpenSSH option: AllowUsers [ TROUVÉ ]  
- OpenSSH option: AllowGroups [ NON TROUVÉ ]  
  
[+] Prise en charge SNMP  
-----
```



```
evenort12@ubuntu:~$ sudo lynis audit system
- net.ipv4.tcp_syncookies (exp: 1) [ OK ]
- net.ipv4.tcp_timestamps (exp: 0 1) [ OK ]
- net.ipv6.conf.all.accept_redirects (exp: 0) [ OK ]
- net.ipv6.conf.all.accept_source_route (exp: 0) [ OK ]
- net.ipv6.conf.default.accept_redirects (exp: 0) [ OK ]
- net.ipv6.conf.default.accept_source_route (exp: 0) [ OK ]

[+] Hardening
-----
- Installed compiler(s) [ NON TROUVÉ ]
- Installed malware scanner [ NON TROUVÉ ]
- Non-native binary formats [ TROUVÉ ]

[+] Tests personnalisés
-----
- Running custom tests... [ AUCUN ]

[+] Plugins (phase 2)
-----

=====

-[ Lynis 3.0.9 Results ]-

Warnings (3):
-----
| Reboot of system is most likely needed [KRNL-5030]
| Solution : reboot
| https://cisofy.com/lynis/controls/KRNL-5030/

| Found some information disclosure in SMTP banner (OS or software name) [MAIL-8818]
| https://cisofy.com/lynis/controls/MAIL-8818/

| No AIDE database was found, needed for AIDE functionality [FINT-4316]
| https://cisofy.com/lynis/controls/FINT-4316/

Suggestions (47):
-----
* This release is more than 4 months old. Check the website or GitHub to see if there is an update available. [LYNIS]
| https://cisofy.com/lynis/controls/LYNIS/

* Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions [DEB-0280]
| https://cisofy.com/lynis/controls/DEB-0280/

=====

evenort12@ubuntu:~$ sudo lynis audit system
* If not required, consider explicit disabling of core dump in /etc/security/limits.conf file [KRNL-5020]
| https://cisofy.com/lynis/controls/KRNL-5020/

* Check PAM configuration, add rounds if applicable and expire passwords to encrypt with new values [AUTH-9229]
| https://cisofy.com/lynis/controls/AUTH-9229/

* Configure password hashing rounds in /etc/login.defs [AUTH-9230]
| https://cisofy.com/lynis/controls/AUTH-9230/

* Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc [AUTH-9262]
| https://cisofy.com/lynis/controls/AUTH-9262/

* When possible set expire dates for all password protected accounts [AUTH-9282]
| https://cisofy.com/lynis/controls/AUTH-9282/

* Configure minimum password age in /etc/login.defs [AUTH-9286]
| https://cisofy.com/lynis/controls/AUTH-9286/

* Configure maximum password age in /etc/login.defs [AUTH-9286]
| https://cisofy.com/lynis/controls/AUTH-9286/

* Default umask in /etc/login.defs could be more strict like 027 [AUTH-9328]
| https://cisofy.com/lynis/controls/AUTH-9328/

* To decrease the impact of a full /home file system, place /home on a separate partition [FILE-6310]
| https://cisofy.com/lynis/controls/FILE-6310/

* To decrease the impact of a full /tmp file system, place /tmp on a separate partition [FILE-6310]
| https://cisofy.com/lynis/controls/FILE-6310/

* To decrease the impact of a full /var file system, place /var on a separate partition [FILE-6310]
| https://cisofy.com/lynis/controls/FILE-6310/

* Disable drivers like USB storage when not used, to prevent unauthorized storage or data theft [USB-1000]
| https://cisofy.com/lynis/controls/USB-1000/

* Check DNS configuration for the dns domain name [NAME-4020]
| https://cisofy.com/lynis/controls/NAME-4020/

* Add the IP name and FQDN to /etc/hosts for proper name resolving [NAME-4404]
| https://cisofy.com/lynis/controls/NAME-4404/

* Install debsums utility for the verification of packages with known good database. [PKGS-7370]
```

```
evenort12@ubuntu:~$ sudo lynis --audit --report /var/log/lynis-report.dat
* You are advised to hide the mail_name (option: smtpd_banner) from your postfix configuration. Use postconf -e or change your main.cf file (/etc/postfix/main.cf) [MAIL-8818]
https://cisofy.com/lynis/controls/MAIL-8818/

* Disable the 'VRFY' command [MAIL-8820:disable_vrfy_command]
- Details : disable_vrfy_command=no
- Solution : run postconf -e disable_vrfy_command=yes to change the value
https://cisofy.com/lynis/controls/MAIL-8820/

* Check iptables rules to see which rules are currently not used [FIRE-4513]
https://cisofy.com/lynis/controls/FIRE-4513/

* Consider hardening SSH configuration [SSH-7408]
- Details : AllowTcpForwarding (set YES to NO)
https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
- Details : ClientAliveCountMax (set 3 to 2)
https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
- Details : LogLevel (set INFO to VERBOSE)
https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
- Details : MaxSessions (set 10 to 2)
https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
- Details : TCPKeepAlive (set YES to NO)
https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
- Details : AllowAgentForwarding (set YES to NO)
https://cisofy.com/lynis/controls/SSH-7408/

* Enable logging to an external logging host for archiving purposes and additional protection [LOGG-2154]
https://cisofy.com/lynis/controls/LOGG-2154/

* Check what deleted files are still in use and why. [LOGG-2190]
https://cisofy.com/lynis/controls/LOGG-2190/

* Add a legal banner to /etc/issue, to warn unauthorized users [BANN-7126]

Lynis security scan details:

Hardening index : 67 [##### ]
Tests performed : 263
Plugins enabled : 1

Components:
- Firewall [ ]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [ ]
- Vulnerability scan [ ]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOFY - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all settings)

evenort12@ubuntu:~$
```

Figure 33 à 48: Ces images attestent l'exécution d'un audit complet du système avec **Lynis**, afin d'obtenir le **score final** de sécurité du serveur (voir pages 22 à 29).

Comparer :

ancien score et nouveau score :

a) interprétation:

Initial : 63 et Final : 67

Le score du côté final est plus élevé que celui du côté initial, ce qui montre que le serveur ou l'audit effectué à final applique davantage de mesures de sécurité. L'augmentation de 63 → 67 indique une amélioration de la configuration de sécurité.

b) Tests effectués (Tests performed):

Initial: 257 et Final: 263

Plus de tests ont été réalisés lors de l'audit de droite. Cela peut expliquer l'augmentation de **l'Hardening Index**, car davantage de contrôles ont été vérifiés.

Amélioration de la sécurité du serveur

13. Installer le service

`sudo apt install unattended-upgrades -y`

13.1 Configurer : `sudo dpkg-reconfigure unattended-upgrades`

Choisir : Yes

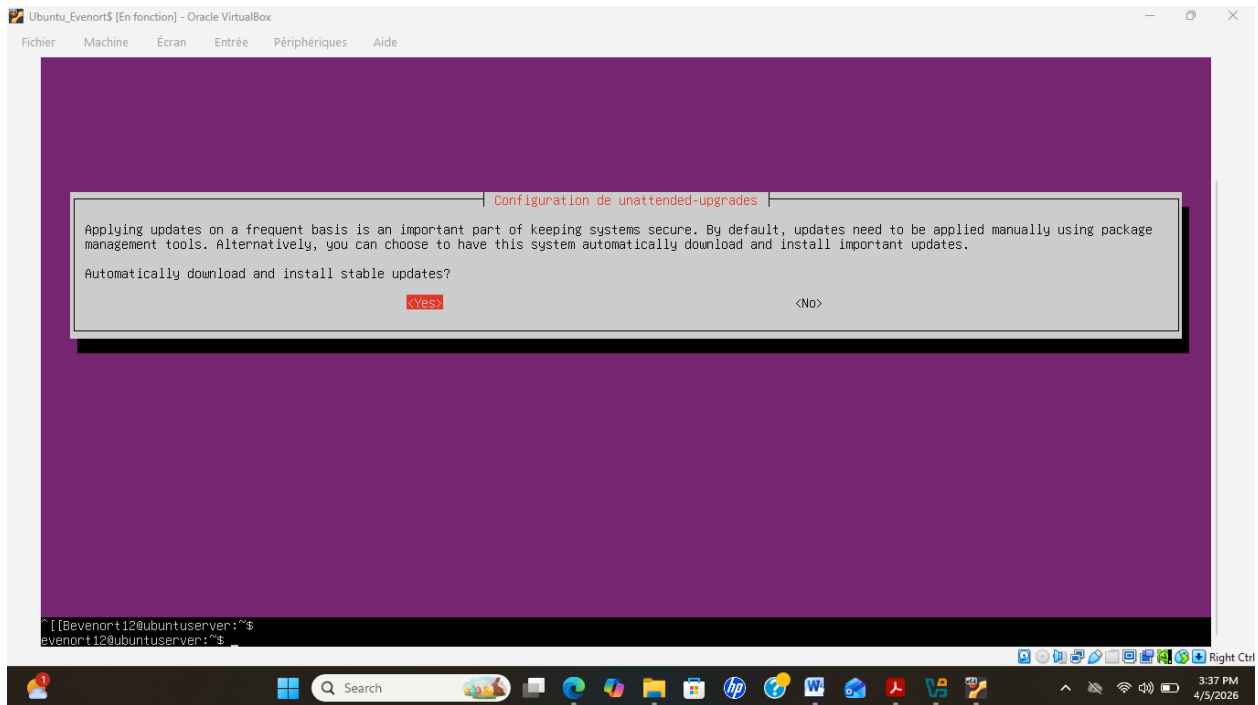


Figure 49: Cette image illustre la configuration avec la commande `sudo dpkg-reconfigure unattended-upgrades` afin d'améliorer la sécurité du serveur.

13.2 Vérifier le fonctionnement :

`systemctl status unattended-upgrades`

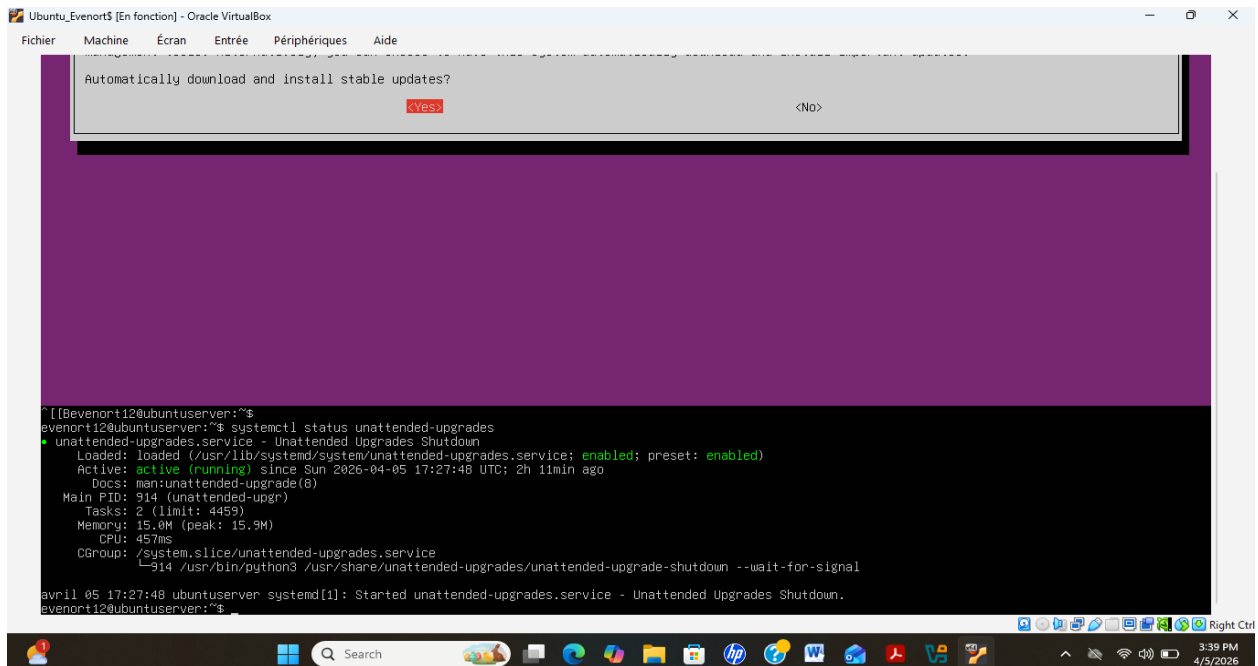


Figure 50 : Cette image illustre la vérification du fonctionnement.

Installer un système de détection d'intégrité

14. Installer : `sudo apt install aide -y`

14.1 Initialiser la base de données : `sudo aideinit`

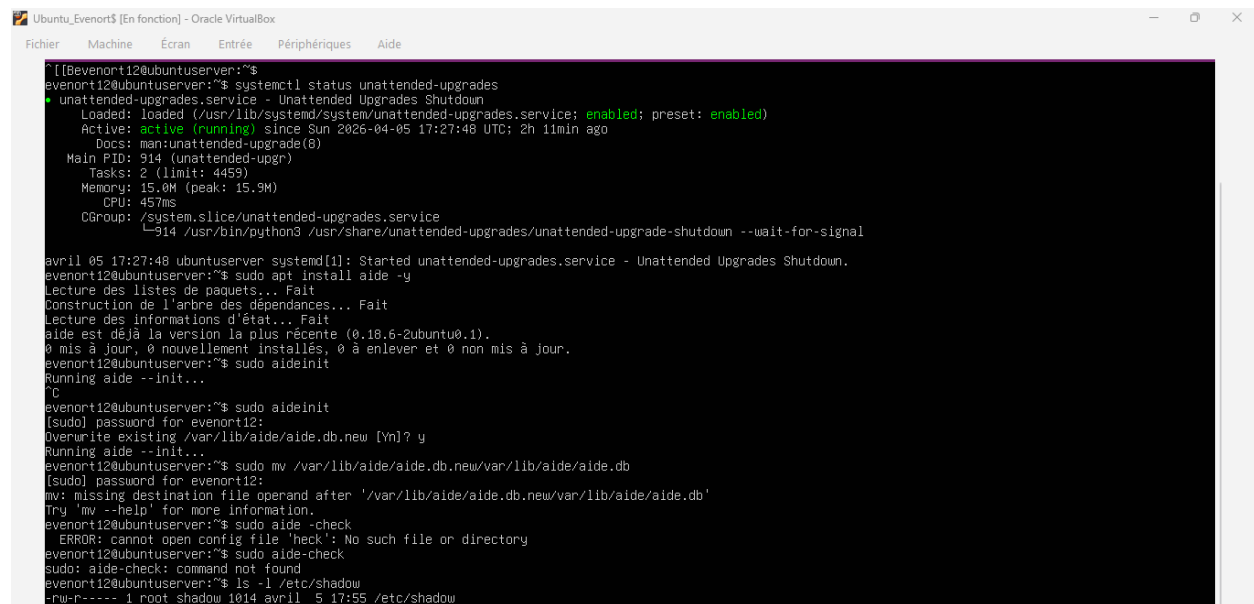
14.2 Déplacer la base : `sudo mv /var/lib/aide/aide.db.new /var/lib/aide/aide.db`

14.3 Lancer un premier scan : `sudo aide --check`

Renforcer les permissions des fichiers sensibles

15. Vérifier les permissions du fichier shadow

`ls -l /etc/shadow`



```
Ubuntu_Evenort$ [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

[[Bevenort12@ubuntu:~]$
evenort12@ubuntu:~$ systemctl status unattended-upgrades
• unattended-upgrades.service - Unattended Upgrades Shutdown
   Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-04-05 17:27:48 UTC; 2h 11min ago
     Docs: man:unattended-upgrade(8)
   Main PID: 914 (unattended-upgr)
    Tasks: 2 (limit: 4459)
   Memory: 15.0M (peak: 15.9M)
      CPU: 457ms
   CGroup: /system.slice/unattended-upgrades.service
           └─914 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal

avril 05 17:27:48 ubuntu:systemd[1]: Started unattended-upgrades.service - Unattended Upgrades Shutdown.
evenort12@ubuntu:~$ sudo apt install aide -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
aide est déjà la version la plus récente (0.18.6-2ubuntu0.1).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
evenort12@ubuntu:~$ sudo aideinit
Running aide --init...
^C
evenort12@ubuntu:~$ sudo aideinit
[sudo] password for evenort12:
Overwrite existing /var/lib/aide/aide.db.new [Yn]? y
Running aide --init...
evenort12@ubuntu:~$ sudo mv /var/lib/aide/aide.db.new/var/lib/aide/aide.db
[sudo] password for evenort12:
mv: missing destination file operand after '/var/lib/aide/aide.db.new/var/lib/aide/aide.db'
Try 'mv --help' for more information.
evenort12@ubuntu:~$ sudo aide --check
ERROR: cannot open config file 'heck': No such file or directory
evenort12@ubuntu:~$ sudo aide-check
sudo: aide-check: command not found
evenort12@ubuntu:~$ ls -l /etc/shadow
-rw-r----- 1 root shadow 1014 avril  5 17:55 /etc/shadow
```

Figure 52 : Sur cette image, l'installation du **package aide** a été réalisée, ainsi que l'initialisation de la base de données, en déplaçant celle-ci.

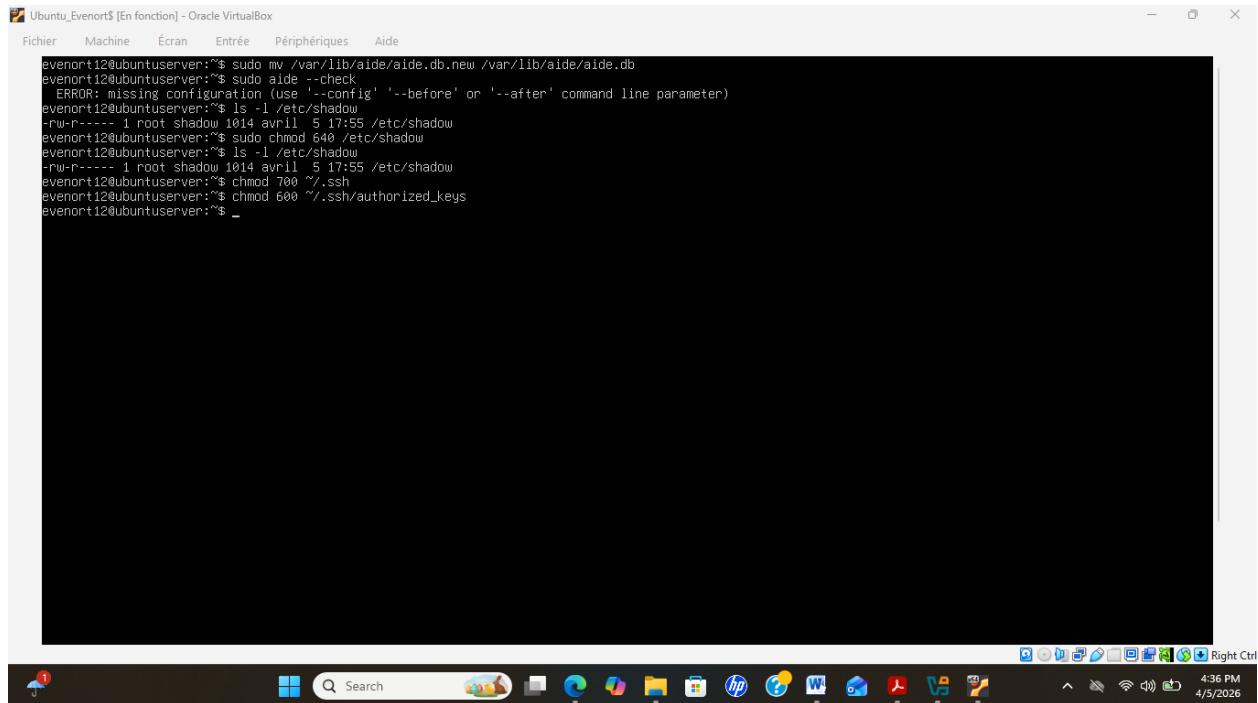
15.1 Corriger si nécessaire : `sudo chmod 640 /etc/shadow`

15.2 Vérifier les permissions du dossier SSH : `ls -ld ~/.ssh`

- **Corriger :**

`chmod 700 ~/.ssh`

`chmod 600 ~/.ssh/authorized_keys`



```
evenort12@ubuntu:~$ sudo mv /var/lib/aide/aide.db.new /var/lib/aide/aide.db
evenort12@ubuntu:~$ sudo aide --check
ERROR: missing configuration (use '--config' '--before' or '--after' command line parameter)
evenort12@ubuntu:~$ ls -l /etc/shadow
-rw-r----- 1 root shadow 1014 avril  5 17:55 /etc/shadow
evenort12@ubuntu:~$ sudo chmod 640 /etc/shadow
evenort12@ubuntu:~$ ls -l /etc/shadow
-rw-r----- 1 root shadow 1014 avril  5 17:55 /etc/shadow
evenort12@ubuntu:~$ chmod 700 ~/.ssh
evenort12@ubuntu:~$ chmod 600 ~/.ssh/authorized_keys
evenort12@ubuntu:~$ _
```

Figure 53: Sur cette image, les permissions ont été vérifiées en consultant les permissions du dossier SSH.

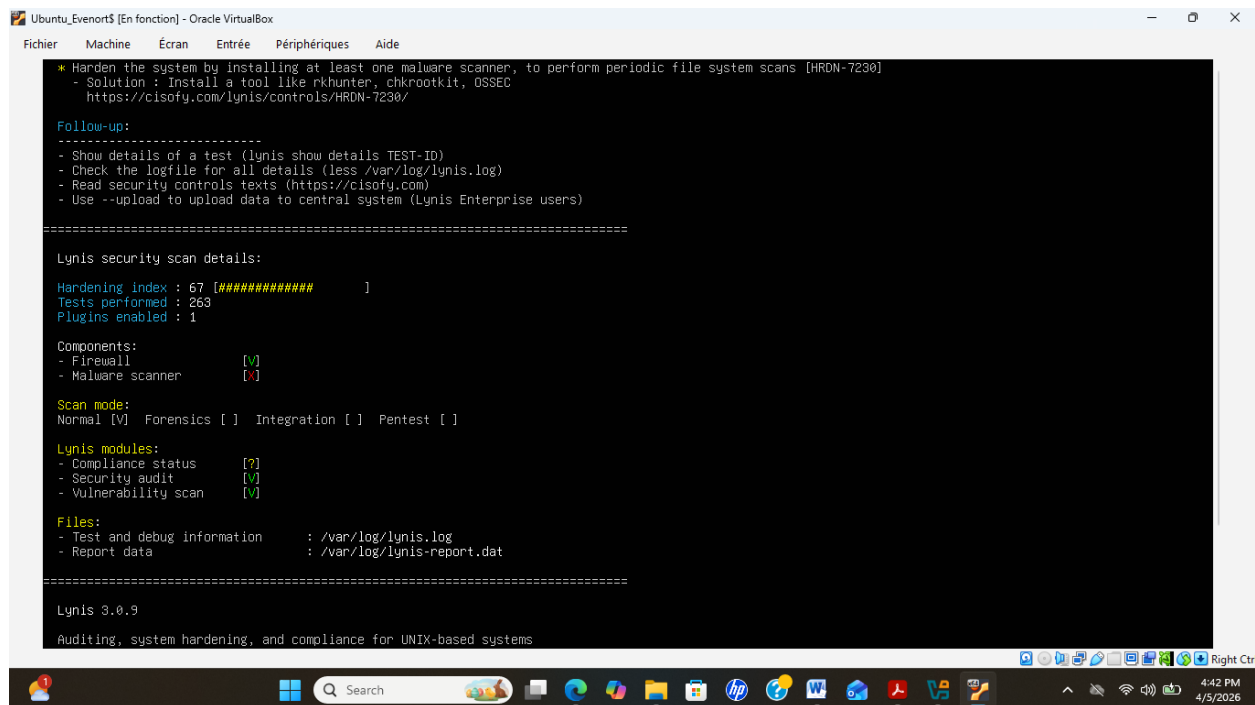
15.3 Vérifier l'espace disque et les logs

15.4 Installer un outil de rotation : `sudo apt install logrotate`

Relancer l'audit de sécurité

15.6 Après les améliorations, relancer l'audit :

`sudo lynis audit system`



```
* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]
- Solution : Install a tool like rkhunter, chkrootkit, OSSEC
https://cisofy.com/lynis/controls/HRDN-7230/

Follow-up:
-----
- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)
=====

Lynis security scan details:

Hardening index : 67 [##### ]
Tests performed : 263
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat
=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
```

Comparer :

ancien score et nouveau score :

a) interprétation:

Initial : 67 et Final : 67

Le score final est identique au score initial, ce qui indique que la configuration de sécurité est restée stable entre les deux audits. Aucune amélioration ni détérioration significative n'a été constatée.

b) Tests effectués (Tests performed):

Initial: 263 et Final: 263

Le nombre de tests réalisés est le même pour les deux audits. L'Hardening Index n'a donc pas changé, ce qui suggère que le niveau de vérification des contrôles est resté constant.

Conclusion

En conclusion, l'audit de sécurité réalisé avec Lynis a permis d'identifier plusieurs points faibles du serveur et d'appliquer des mesures de sécurité comme les mises à jour automatiques, la configuration du firewall, l'installation d'un IDS comme AIDE et la sécurisation des permissions des fichiers système. Après ces améliorations, le serveur est plus sécurisé, mais il est important de comprendre qu'aucun système n'est totalement sécurisé. La sécurité informatique est un processus continu qui nécessite une surveillance régulière, des mises à jour et des audits fréquents.

Pourquoi les mises à jour automatiques sont-elles importantes en production ?

Les mises à jour automatiques sont importantes en production parce qu'elles permettent de corriger rapidement les failles de sécurité découvertes dans le système ou les logiciels installés.

Un serveur non mis à jour peut être vulnérable aux attaques (exploits, malware, accès non autorisé).

Les mises à jour automatiques permettent aussi d'améliorer la stabilité, les performances et la sécurité du serveur sans intervention manuelle. En outre Elles permettent de protéger le serveur contre les nouvelles vulnérabilités.

2. Quel est le rôle d'un IDS comme AIDE ?

Un IDS (Intrusion Detection System) comme **AIDE** sert à détecter les intrusions sur le système.

AIDE vérifie l'intégrité des fichiers importants du système (ex : /etc, /bin, /usr).

Si un fichier est modifié, supprimé ou ajouté sans autorisation, AIDE envoie une alerte. AIDE permet de détecter les modifications suspectes sur le serveur.

3. Pourquoi sécuriser les permissions des fichiers système ?

Sécuriser les permissions des fichiers système permet d'empêcher les utilisateurs non autorisés de modifier les fichiers critiques du système.

Si les permissions sont mal configurées, un attaquant peut :

- modifier la configuration du système
- ajouter un virus
- créer un accès administrateur
- supprimer des fichiers importants

Conclusion : Les bonnes permissions protègent l'intégrité et la sécurité du système.

4. Peut-on considérer un serveur totalement sécurisé ?

Non, un serveur ne peut jamais être totalement sécurisé.

Il existe toujours de nouvelles failles, de nouvelles techniques d'attaque et des erreurs humaines possibles.

On ne peut pas avoir une sécurité à 100%, mais on peut réduire les risques en :

- mettant à jour le système
- configurant le firewall
- utilisant Fail2ban
- utilisant AIDE
- désactivant les services inutiles
- surveillant les logs
- faisant des audits avec Lynis

En effet, La sécurité est un processus continu, pas un état final.